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GENERAL APTITUDE

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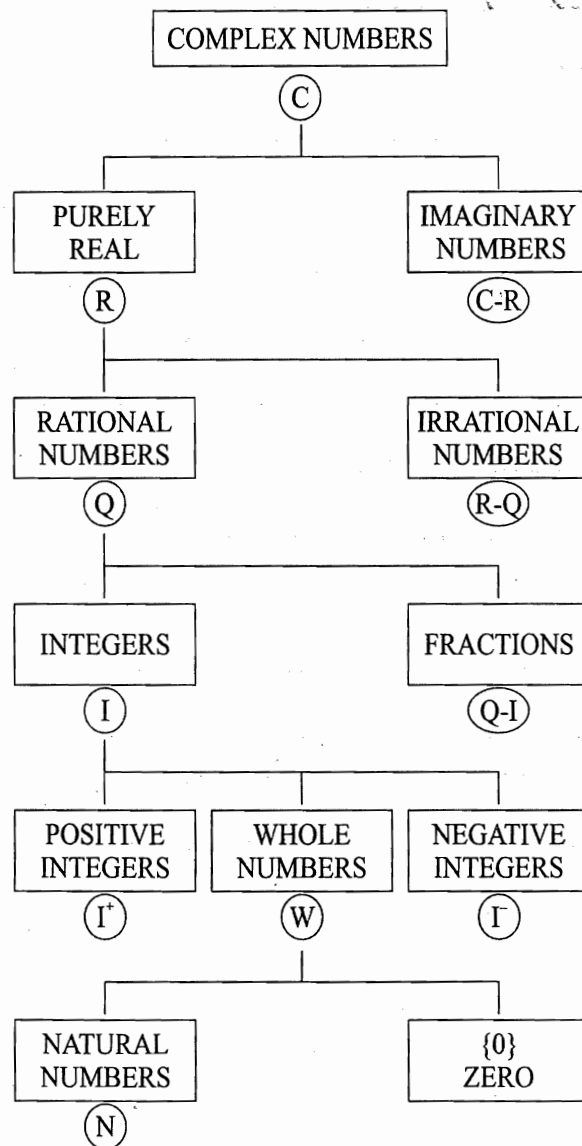
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CHAPTER 1

NUMBER SYSTEM

Basic Introduction



Prime Number: Except 1 each natural number which divisible by only 1 and itself is called as prime number e.g.

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,etc.

- There are total 25 prime number upto 100
- There are total 46 prime number upto 200

- 2 is the only even prime number and the least prime number.
- 1 is neither prime nor composite number
- There are infinite prime numbers.
- A list of all the prime number upto 100 is given below.

Table of Prime Number (1-100):

2	11	23	31	41	53	61	71	83	97
3	13	29	37	43	59	67	73	89	
5	17		47			79			
7	19								

How to test whether a number is prime or not: To test a number n take the square root of n and consider as it is, if it is a natural number otherwise just increase the square root of it to the next natural number. Then divide the given number by all the prime numbers below the square root obtained. If the number is divisible by any of these prime numbers then it is not a prime number else it is a prime number.

Example: Check that whether 241 is prime.

Solution: When we take the square root of 241 it is approximate 15, so we consider it 16. Now we divide 241 by all the prime numbers below 16 viz., 2, 3, 5, 11, 13

Since 241 is not divisible by any one of the prime numbers below 16. So it is a Prime Number.

Examples:

- The product of two integers is 30, where the even integer is greater than the odd integer. What can be their minimum sum.
- Find the all possible solution of $a^b + b^a = 17$, where a, b are natural numbers?
- $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8, \dots + 99 - 100?$
- $7 - 8 + 9 - 10, \dots + 99 - 100?$
- $7 - 8 + 9 - 10, \dots + 1699 - 1700 + 1701?$
- $-8 + 9 - 10, \dots + 99 - 100?$
- If $a \times b \times c \times d \times e = 45$ where, a, b, c, d, e are distinct integers then find $a + b + c + d + e = ?$
- There are 15 students in a class room. Each one shakes hand with one another exactly once. Find the number of total hand shakes occurred in the class?

Put Your Own Notes

- (i) On the occasion of New Year every student of a class exchanged greeting card with every other student. If total 870 cards have been exchanged, find the total number of students in the class?
- (j) The average of any 5 consecutive natural numbers is x . If next number is also included, the average becomes y . Find $y - x$.
- (k) If $a^3 + b^3 = 217$ and $a^3 + b^3 + c^3 = d^3$, where $a^3 + b^3 + c^3 = d^3$, where a, b, c, d are distinct natural numbers, find $a + b + c + d$?

1.1 Problems based on $\sum n, \sum n^2, \sum n^3$

- (i) $\sum n, \sum n^2, \sum n^3$
- (ii) Consider the sequence of ordered sets of natural numbers:
- (iii) $\{1\}, \{2, 3\}, \{4, 5, 6\}, \dots$
- (iv) What is the last number in the 10^{th}
- (v) The sum of first ' n ' natural numbers with one of them missed is 42. What is the number that was missed?
- (vi) What is the 94^{th} term of the following sequence?
- (vii) 1, 1, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4
- (viii) A pyramid shaped toy is made by tightly placing cubic blocks of $1 \times 1 \times 1 \text{ cm}^3$. The base of the toy is a square $4 \times 4 \text{ cm}^2$. The width of each step is 0.5 cm. How many blocks are required to make the toy?
- (ix) Find the average of the given series
- (x) 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6, 7, 7, 7, 7, 7, 7, 7
- (xi) If $1^3 + 2^3 + 3^3 + 4^3 + \dots + 10^3 = 3025$,
then find $2^3 + 3^3 + 4^3 + \dots + 20^3 = ?$
- (xii) $xy - yx = 9|x - y|, xy + yx = 11(x + y), \quad xyx - zyx = 99|x - z|$
- (xiii) If we add 54 to a two digit number, its digits get reversed. How many such two digit number are possible?
- (xiv) If we add 18 to a two digit number, its digits get reversed. If the original number and the new number both are prime. How many such numbers exist?
- (xv) The difference between a two digit number and its reverse is 45. How many such number exist?

1.2 Cyclicity (Unit Digit)

- (i) Find the unit digit of $1! + 2! + 3! + \dots + 100! = ?$
- (ii) Find the unit digit of $2^{522}, 7^{71}, 5^{72}$

1.3 Exponents (Number of Zeroes)

- (i) What can be the maximum power of 2 that will completely divide $10!$?
- (ii) Find the number of zeroes at the end of $100!$?
- (iii) If $n!$ end with 14 zeroes, find the sum of all possible values of n ?

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- (iv) If $n!$ end with 30 zeroes, how many values of n are possible?
- (v) If N is the product of first 100 multiples of 10. Then find the number of zeroes at the end of N ?
- (vi) How many digits are there in the following product
 $2^{17} \times 3^2 \times 5^{14} \times 7 = ?$
- (vii) A function $f(n)$ is defined such that $f(n) = \text{sum of digits of } n$. For
 ex. $f(137) = 1 + 3 + 7 = 17$, find $f(2^6 \times 3^5 \times 5^5)$.

1.4 Factors

- (i) How many factors of 400 are perfect squares.
- (ii) Find the total number of factor of 360.
- (iii) Find the total number of factor of 5000.
- (iv) Find the least number with 7 factors.
- (v) Find the least number with 17 factors.
- (vi) Find the least number with 6, 18, 12 factors.
- (vii) If N^2 has 15 factors, how many factors can N have?
- (viii) The number of odd factors of 360, 1000.
- (ix) Find the product of all the factors of 100.
- (x) Find the sum of all the factors of 100.

1.5 Divisibility

- (i) Any four digit number of the form $abab$ is always a multiple of ?
- (ii) Any six digit number of the form $abab$ is always a multiple of ?
- (iii) If $12x456$ is a multiple of 11, find the value of x ?
- (iv) If $327x92$ is a multiple of 11, find the value of x ?
- (v) If $652x38$ is a multiple of 11, find the value of x ?
- (vi) If $75x73$ is a multiple of 12, how many such number are possible x ?
- (vii) Find the common factor of $(11^{15} + 15^{15})$ and $(11^{11} + 15^{11})$.

1.6 Remainder

Find the remainder of the following

$$\frac{13^{25}}{3}, \frac{2^{96}}{17}, \frac{15^{23}}{11}, \frac{4^{64}}{6}, \frac{11^{24}}{3}, \frac{52!}{53}$$

$$\frac{15^{19} + 23^{19}}{19}, \frac{12^{73} + 30 + 24^{73} + 34^{73}}{30}$$

$$\frac{1 + 3^3 + 5^5 + 7^7 + \dots + 51^{51}}{8}$$

$$\frac{3^0 + 3^1 + 3^2 + 3^3 + \dots + 3^{79}}{13}$$

1.7 LCM and HCF

- (i) The HCF of two numbers is 33 and their sum is 528, how many such pairs exist?
- (ii) The HCF of two numbers is 13 and their sum is 2028, how many such pairs exist?
- (iii) The HCF of two numbers is 20 and their LCM is 600, how many such pairs exist?

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CHAPTER 2

TIME AND DISTANCE

The terms time and distance are related to the speed of a moving object.

Speed: It is the distance travels per unit time.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Conversion of units

$$\text{One kilometre / hour} = \frac{1000 \text{ metre}}{60 \times 60 \text{ seconds}} = \frac{5}{18} \text{ m/sec.}$$

Useful Formulae

$$(i) \quad \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$(ii) \quad \text{Distance} = (\text{Speed} \times \text{Time})$$

$$(iii) \quad \times \text{m/sec} = \left(\times \times \frac{18}{5} \right) \text{ km/hr}$$

(iv) If a certain distance is covered at u km/hr and the same distance is covered at v km/hr, then the average speed during whole journey

$$= \frac{2uv}{(u+v)} \text{ km/hr.}$$

(v) If a man changes his speed in the ratio $x : y$, then the ratio of times taken becomes $x : y$.

2.1 Relative Speed

It is the speed of a moving object in relation to other moving object. Relative speed of a moving body with speed $\times \text{km/h}$ in relation to the other moving body with speed $y \text{ km/h}$ in same direction is difference of their speed i.e., $(x - y) \text{ km/h}$ and when the body is moving in opposite direction is the sum of their speeds i.e., $(x + y) \text{ km/h}$.

Useful Formula:

- (i) If two bodies are moving in the same direction, then their relative velocity will be equal to the difference of their speeds. Suppose two trains or bodies are moving in the same direction at u km/hr and v km/hr such that $u > v$, then their relative speed $= (u - v) \text{ km/hr}$. If two trains of lengths $\times \text{km}$ and $y \text{ km}$ move in the same direction at $u \text{ km/hr}$ and $v \text{ km/hr}$ (where $u > v$), then time taken to cross each

other is $\left(\frac{x+y}{u-v}\right)$ hrs. Where x meter and y meter are the length of two trains.

- (ii) If two bodies are moving in the opposite directions then their relative velocity will be equal to the sum of their speeds. Suppose two trains or bodies are moving in the opposite directions at u km/hr and v km/hr. Then, relative speed $= (u+v)$ km/hr. If two trains of lengths x km and y km moved in the opposite directions at u kmph and v kmph, then the time taken to cross each other is $\left(\frac{x+y}{u+v}\right)$ hrs. Where x meter and y meter are the length of two trains.
- (iii) If two persons (or vehicles) start at the same time from two points A and B towards each other and after crossing, they take x and y units of time in reaching B and A respectively, then $\frac{\text{Speed of first}}{\text{Speed of second}} = \sqrt{\frac{y}{x}}$
- (iv) Time taken by $\text{train} \times \text{meters}$ long in passing a signal post or a standing man is the same as the time taken by the train to cover $\times \text{meters}$ with its own speed.
- (v) Time taken by a $\text{train} \times \text{meters}$ long in passing a stationary object of length y meters is the same as the time taken by the train to cover $(x+y)$ meters with its own speed.

2.2 Boats and Streams

Downstream: The direction along the stream (water) is called downstream.

Up stream: The direction against the stream is called upstream.

- (i) Let the speed of a boat in still water be u km/hr and let the speed of the stream be v km/hr. Then:
- (a) Speed downstream $= (u+v)$ km/hr
- (b) Speed upstream $= (u-v)$ km/hr
- (ii) Let speed downstream $= u$ km/hr and speed upstream $= v$ km/hr. Then,
- (a) Rate in still water $= \frac{1}{2}(u+v)$ km/hr
- (b) Rate or current $= \frac{1}{2}(u-v)$ km/hr
- (iii) If a man rows a certain distance downstream in t_1 hours and returns the same distance in t_2 hours, and the stream flows at the rate of

Put Your Own Notes

u km/hr, then speed of the man in still water is $\frac{u(t_1 + t_2)}{t_2 - t_1}$ km/hr.

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Examples:

- (a) A train, when it moves at an average speed of 40 kmph, reaches its destination on time. When its average speed becomes 35 kmph, it reaches its destination 15 minutes late. Find the length of the journey.

Sol: Suppose the destination is D km away from the starting point.

Now, according to the question, $\frac{D}{35} - \frac{D}{40} = 15 \times \frac{1}{60}$ [Convert minute into hour]

$$\Rightarrow \frac{40D - 35D}{35 \times 40} = \frac{15}{60}$$

$$\Rightarrow \frac{5D}{35 \times 40} = \frac{15}{60}$$

$$\Rightarrow D = \frac{15 \times 35 \times 40}{60 \times 5} = 70 \text{ km}$$

$\therefore$ Required length of the journey = 70 km.

- (b) A car covers the first 30 km of its journey in 45 minutes and the remaining 25 km in 35 minutes. What is the average speed of the car?

$$\text{Sol.: Average speed} = \frac{\text{Total distance travelled}}{\text{Time taken to travel the distance}} = \frac{30 + 25}{\left(\frac{45 + 35}{60}\right)}$$

$$= \frac{55 \times 60}{80} = 41\frac{1}{4} \text{ km/hr}$$

CHAPTER 3

TIME AND WORK

Working efficiency is the work done by an individual in one day and this efficiency is inversely proportional to the number of days to complete a work. It means that a person who takes less days to complete a work is said to be more efficient than a man who takes more days to complete the same work. For example, if A can do a piece of work in n days, then the part of the work by him in 1 day $= \frac{1}{n}$.

3.1 Important Facts

- (i) If A's 1 day's work $= \frac{1}{n}$, then the time taken by A to finish the work = n days. $\text{one day's work} = \frac{1}{\text{Number of days required to finish the work}}$
- (ii) The total work is assumed to be 1.
- (iii) If 'A' can do a piece of work in x days 'B' can do it in y days then 'A' and 'B' working together can do the same work in $\frac{xy}{xy + yz + zx}$ days.
- (iv) If A, B and C can do a work in x, y and z days respectively then all of them working together can finish the work in $\frac{xyz}{xy + yz + zx}$ days.
- (v) Time and number of men each are always in direct proportion to work. If M_1 persons can do W_1 works in D_1 $W_2 = M_2 D_2 W_1$.
We can also conclude that
 - (a) More also less days and conversely more days lessd men.
 - (b) More men more work and conversely more work more men.
 - (c) More days more work and conversely more work more days.
- (vi) The time required to complete a certain apiece of work is increased 9or decreased) in the same ratio in which the number of men engaged of do that piece of work is decreased 9or increased).

3.2 Pipes and Cisterns

Important Facts:

- (i) If an inlet pipe can fill a cistern in x hours, then the part filled in 1 hour $= \frac{1}{x}$.
- (ii) If a pipe can empty a full tank in y hours, then part of tank emptied in

$$1 \text{ hour} = \frac{1}{y}$$

- (iii) If an inlet pipe can fill a tank in x hours and an outlet pipe empties the full tank in y hours, then the net part filled in 1 hour when both the

pipes are opened $= \left(\frac{1}{x} - \frac{1}{y} \right)$ (where $y > x$).

- (iv) If an inlet pipe can fill a tank in x hours and an outlet pipe empties the

full tank in hours, then net part emptied in 1 hour $= \left(\frac{1}{y} - \frac{1}{x} \right)$ (where $x > y$).

Examples:

- (a) The work done by a child in one day is one third that by a man half that by a woman. If one man, one woman and one child together can complete a work in 2 days, in how days can 4 children together complete the same work?

Sol: We have, $1M = 3C$ and $1W = 2C$ (where, M = male work, C = child work, W = woman work) Therefore, $1M + 1W + 1C = 6C$

Hence, the required number of days $= \frac{6 \times 2}{4} = 3 \text{ days}$

- (b) A tank is filled in 5 hours by three pipes A, B and C. The pipe C is twice as fast as B and B is twice as fast as A. How much time will pipe A alone take to fill the tank?

Sol: Given, ratio of efficiencies of pipes A, B and C are as follows

C	B	A
2	1	
	2	1
4	: 2	: 1

Suppose the efficiencies of pipes C, B and A are $4K$, $2K$ and K respectively.

Since, the tank is filled in 5 hours by the three pipes having combined efficiency equal to $7k$ then the time required to fill the tank by A alone

$$= \frac{7k \times 5}{k} = 35 \text{ hours}$$

- (c) A and B together can dig a trench in 12 days, which A alone can dig in 30 days. In how many days B alone can dig it?

Sol: $(A + B)$'s 1 day's work $= \frac{1}{12}$

$$A's \text{ 1 day's work} = \frac{1}{30}$$

$$\therefore B's \text{ 1 day's work} = \frac{1}{12} - \frac{1}{30} = \frac{5-2}{60} = \frac{3}{60} = \frac{1}{20}$$

Hence, B alone can dig the trench in 20 days.

Put Your Own Notes

CHAPTER 4

SCHEDULE DAY/DATE/TIME

In a normal year there are 365 days i. e. , 52 weeks and 1 day. Therefore, a normal year contains 1 odd day. Leap year: A leap year has 366 days.

- (i) Every year divisible by 4 is a leap year, if it is not a century
- (ii) Every year divisible by 4 is a leap no other century is a-leap year.

4.1 Counting of Odd Days

- (i) 1 ordinary year = 365 days = (52 weeks + 1 day.)

$\therefore$ 1 ordinary year has 1 odd day.

- (ii) 1 leap year = 366 days = (52 weeks + 2 days)

1 leap year has 2 odd days.

- (iii) 100 year = 76 ordinary years + 24 leap years

$= (76 \times 1 + 24 \times 2)$ odd = days = 124 odd days.

$= (17 \text{ weeks} + 5 \text{ days}) = 5$ odd days.

Number of odd days in 100 years = 5

Number of odd days in 200 years $= (5 \times 2) = 3$ odd days (Here, we divide 10 days by 7, remainder is 3).

Number of odd days in 300 years $= (5 \times 3) = 11$ odd day

Number of odd days in 400 years $= (5 \times 4 + 1) = 0$ day.

Similarly, each one of 800 years, 1200 years, 1600 years, 2000 years etc. Has an odd day.

4.2 Day of the Week Related to Odd Days

In a given period, the number of days more than the complete weeks are called odd days.

Number of days	0	1	2	3	4	5	6
Day	Sun	Mon	Tues	Wed	Thurs	Fri	Sat

4.3 Clocks

The dial of a clock is a circle whose circumference is divided into 12 parts, called hour spaces. Each hour space is further divided into 5 parts, called minute spaces. This way, the whole circumference is divided into $12 \times 5 = 60$ minute spaces.

The time taken by the hour hand (Smaller hand) to cover a distance of an hour space is equal to the time taken by the minute hand (longer hand) to cover a distance of the whole circumference. Thus, we may conclude that in 60 minutes, the minute-hand gains 55 minutes over the hour-hand.

In an hour, the hour-hand moves a distance of 5 minute spaces whereas the

minute-hand moves a distance of 60 minute spaces. Thus, the minute-hand remains $60 - 5 = 55$ minute spaces ahead of the hour-hand.

Important Facts:

- (i) In every hour, both the hands coincide once.
- (ii) When the two hands are at right angle, they are 15 minute spaces apart. This happens twice in every hour.
- (iii) When the hands are in opposite directions, they are 30 minute spaces apart. This happens once in every hour.
- (iv) The hour hand moves around the whole circumference of clock once in 12 hours. So the minute-hand is twelve times faster than the hour-hand.
- (v) The clock is divided into 60 equal minute divisions.
- (vi) 1 minute division = $360^\circ / 60 = 6^\circ$ apart.
- (vii) Each hour number is evenly and equally separated by five minute divisions ($5 \times 6^\circ = 30^\circ$ apart).
- (viii) In one minute, the hour hand moves $\frac{1^\circ}{2}$.
- (ix) In one minute the minute-hand gains $5\frac{1^\circ}{2}$ more than the hour-hand.
- (x) A clock will subtend an angle 6° between them of the two hands of a clock. At any time, the same can be found out using the following formula for the variants in a clock:

$$\theta = \frac{11}{2}m - 30h \left(\text{when } \frac{11}{2}m > 30h \right)$$

Or

$$\theta = 30h - \frac{11}{2}m \left(\text{when } 30h > \frac{11}{2}m \right)$$

Examples:

- (a) It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2010?

Sol: On 31st December 2005, it was Saturday.

Number of odd days from the year 2006 to the year 2009 = $(1 + 1 + 2 + 1) = 5$ days.

$\therefore$ On 31st December 2009, it was Thursday.

Thus, on 1st Jan, 2010 it is Friday.

- (b) A watch which gains uniformly is 2 minutes low at noon on Monday and is 4 min. 48 sec fast at 2 p.m. on the following Monday. When was it correct?

Sol: Time from 12 p.m. on Monday to 2 p.m. on the following Monday = 7 days 2 hours = 170 hours.

$\therefore$ The watch gains $\left(2 + 4\frac{4}{5} \right) \text{ min} = \frac{34}{5} \text{ min}$ in 170 hrs.

Now, $\frac{34}{5} \text{ min}$. Are gained in 170 hrs.

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2min. Are gained in $\left(170 \times \frac{5}{34} \times 2\right)$ hrs = 50 hrs.

Therefore, watch is correct 2 days 2 hrs. After 12 p.m. on Monday i. e. , it will be correct at 2 p.m. on Wednesday.

- (c) A clock loses 1 % time during the first week and then gains 2 % time during the next one week. If the clock was set right at 12 noon on a Sunday, What will be the time that the clock will show exactly 14 days from the time it was set right?

Sol: the clock loses 1% time during the first week.

In a day there are 24 hours and in a week there are 7 days. Therefore, there are $7 \times 24 = 168$ hours in a week.

If the clock loses 1% time during the first week, then it will show a time which is 1% of 168 hours less than 12 Noon at the end of the first week = 1.68 hours less.

Subsequently, the clock gains 2% during the next week. The second has 168 hours and the clock gains 2% time = 2% of 168 hours = 3.36 hours more than the actual time.

As it lost 1.68 hours during the first week and then gained 3.36 hours during the next week, the net result will be a $-1.68 + 3.36 = 1.68$ hour net gain in time. So the clock will show a time which is 1.68 hours more than 12 Noon two weeks from the time it was set right. 1.68 hours = 1 hours = 1 hour and 40.8 minutes = 1 hour + 40 minutes + 48 seconds. That is 1: 40: 48 P.M.

- (d) If the third Sunday of a particular month falls on 20th day of the month, what will be the first day of that month?

Sol: Given, third Sunday = 20th day of the month.

Second Sunday = $20 - 7 = 13^{\text{th}}$ day of the month

First Sunday = $13 - 7 = 6^{\text{th}}$ day of the month

Now, 5th day = Saturday, 4th day = Friday, 3rd day = Thursday, 2nd day = Wednesday, 1st = Tuesday.

- (e) Sumedha would like to complete all her homework before 10 p.m. in order to watch important final cricket match between India and Pakistan on TV. She has 40 minutes assignment in each of her five prepared subjects. What is the latest time at which she can start and still complete her homework in time for the final cricket match?

Sol: Time needed for Sumedha to complete all her 5 assignments = (40×5) minutes = 200 minutes

= 3 hours 20 minutes

Hence, the required time is 3 hours 20 minutes before 10 p.m. so, she
can start work at 6.40 p.m.

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Dips Academy

CHAPTER 5

PROFIT AND LOSS

5.1 Based on CP and SP

- (i) If the cost price is Rs. 60 and profit percent is 20%. Find out the selling price.
- (ii) Find out the selling price, if cost price is Rs. 80 and loss percent is 15%.
- (iii) Find out the cost price, if selling price is Rs. 80 and profit percent is 25%.
- (iv) A man bought some toffees at 3 for a rupee. How many for a rupee should he sell to gain 50% in the entire transaction?
- (v) A man bought some toffees at 6 for a rupee. How many for a rupee should he sell to gain 25% in the entire transaction?
- (vi) Dinesh purchased some lemons. He purchased 3 lemons for Rs. 2. He sold all the lemons and earned 25% profit. At
Ans: 6 for Rs. 5
- (vii) Ratish purchased some lemons. He sold all the lemons. He sold 8 lemons and earned 25% profit. At what rate did he sell all the lemons?
Ans: 16 for Rs. 15
- (viii) Keshav purchased some lemons. He sold all the lemons. He sold 11 lemons for Rs. 10 and incurred 30% loss. What should be the selling rate if he incurs 20% loss?
Ans: 77 for Rs. 80
- (ix) By selling 12 toffees for a rupee a man lost 20%. How many for a rupee should he sell to gain 20%?
- (x) A man sold 50 toffees for a rupee and hence lost 25%. How many for a rupee should he sell to gain 25%?
- (xi) By selling 45 lemons for Rs. 40, a man 20%. How many lemons should he sell for Rs. 24 to gain 20%.
- (xii) A man bought 6 toffees for Rs. 10 and sold them 4 for Rs. 6. Find out his gain or loss percentage.
- (xiii) Mukesh purchased some lemons. He purchased 7 lemons for Rs. 9. He sold all the lemons. If he sold 11 lemons for Rs. 13. What is the percent loss in the whole transaction?
Ans: Approx 8% loss
- (xiv) If the selling price of 50 articles is equal to the cost price of 40 articles, then the loss or gain percent is
- (xv) The C.P. of 21 articles is equal to S.P. of 18 articles. Find the gain of loss percent.

- (xvi) By selling 48 meters of cloth a person gains the cost price of 12 meters. Find the gain per cent.

Ans: 25%

5.2 Defrauding

- (i) A shopkeeper claims to sell his goods at cost price but he uses a false balance which show 1 kg instead of 800 grams. Find out his gain percentage.
- (ii) A shopkeeper marks up his goods by 20% and uses a false balance which show 1 kg instead of 900 grams. Find out his profit percentage.
- (iii) A grocer sell rice at a profit of 20% and uses a weight which is 25% less. Find his total percentage gain.

Ans: 60%

- (iv) A tradesman defrauds to the extent of 10% in buying goods and also defrauds to the extent of 10% in selling. His gain per cent is :

Ans: $22\frac{2}{9}\%$ gain

5.3 When the Profit/ Loss is Calculated on S.P

- (i) By mistake a shopkeeper calculates his profit on selling price and claims to make a profit of 20%. Find his actual profit?
- (ii) A shopkeeper calculates his loss on selling price by mistake and claims a loss of 25%, find his actual profit?

$$x^2 / 100\%$$

- (iii) A man sold two flats for Rs. 675958 each. On one he gains 16% while on the other he loss 16%. How much does he gain or loss in the whole transaction?
- (iv) A man sold two articles, each for the same price Rs. 600. He earns 20% profit on the first and incurs 20% loss on the second. What would be his overall profit or loss ? (in Rs)

5.4 Unitary Method

- (i) A man sold his watch at 5% loss, if the selling price had increased by Rs. 2000, there would have been a gain of 15%. Find out the cost price of the watch?
- (ii) A man sells an article at a profit of 25%. If he had bought it at 20% less and sold it for Rs. 10.50 less, he would have gained 30%. Find the cost price of the article.

5.5 Reverse Ration

A person bought two watches for 960. He sold one at a loss of 20% and the other at a gain of 60% and the found that each watch was sold at the same price. Find the cost price of two watches.

Ans: Rs 640, Rs 320

5.6 Difference Division Method

The profit earned by selling an article for Rs 832 is equal to the loss incurred when the same article is sold for Rs. 448. What should be the sale price of the article for making 50 percent profit?

Ans: Rs. 960

5.7 Miscellaneous

- (i) A trader gives 4% discount on the MRP of a pen and gives one pen free for buying every 15 pens, and hence he makes 35% profit of the entire transaction. The MRP which is above the cost price is?
- (ii) A dealer sold three-fourth of his articles at a gain of 20% and the remaining at cost price. Find the gain earned by him in the whole transaction.
- (iii) A tradesman marks his goods 20% above the cost price. He offers 10% discount to customers. What is his gain in per cent?

Ans: 8% gain

- (iv) Radha purchased 10 dozens pens at Rs. 70 per dozen. He sold 7 dozens pen at 7% profit and the remaining 3 dozens at 17% profit. What is his profit percentage in this transaction?

Ans: 10%

- (v) 20 kg of potato costs as much as 5 kg of tomato, 12 kg tomato cost as much as 30 kg of onion, 15 kg of onion costs as much as 18 kg of cabbage. If 10 kg of cabbage costs Rs. 50. What would be the cost of 24 kg of potato?

Ans: Rs 90

- (vi) Two third of a commodity is sold at 30% profit, one fourth is sold at 16% profit and the remaining at 12% profit. If a total profit of Rs. 100 is earned, then find the value of the commodity.

Ans: Rs.400

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CHAPTER 6

PERMUTATIONS AND COMBINATION

Before discussing Permutations and Combinations, let us look at what is called as the "Fundamental Rule".

"If one operation can be performed in 'm' ways and (when, it has been performed in any one of these ways), a second operation then can be performed in 'n' ways, the number of ways of performing the two operations will be $m \times n$ ".

This can be extended to any number of operations.

If there are three cities A, B and C such that there are 3 roads, connecting A and B, and 4 roads connecting B and C, then the number of ways one can travel from A to C via B is 3×4 , i.e., 12.

6.1 Permutations

Each of the arrangements which can be made by taking some or all of a number of items is called a Permutation. Permutation implies "arrangement", that their order of the items is important.

The permutations of three items a, b and c taken two at a time are ab, ba, ac, ca, cb and bc. Since the order in which the items are taken is important, ab and ba are counted as two different permutations. The words "permutation" and "arrangement" are synonymous and can be used interchangeably.

The number of permutations of n things taking r at a time is denoted by nPr (and read as " nPr ").

6.2 Combinations

Each of the groups or selections which can be made by taking some or all of a number of items is called a Combination. In combinations, the order in which the items are taken is not considered as long as the specific things are included.

The combinations of three items a, b and c taken two at a time are ab, bc and ca. Here, ab and ba are not considered separately because the order in which a and b are taken is not important but it is only required that a combination including a and b be counted. The words "combination" and "selection" are synonymous.

The number of combination of n thing taking r at a time is denoted by nC_r
 (and read as " nCr ")

6.3 Number of Linear Permutation of ' n ' Dissimilar Items Taken ' r ' at a Time without Repetition (nP_r)

Consider r boxes each of which can hold one item. When all the r boxed are filled, what we have is an arrangement of r items taken from the given n items. So, each time we fill up the r boxed with items taken from the given n items, we have an arrangement of r items taken from the given n items without repetition. Hence the number of ways in which we can fill up the r boxed by taking things from the given n things is equal to the number of permutations of n things taking r at a time.

Boxes $\square$ $\square$ $\square$ $\square$ $\square$
 1 2 3 4 r

The first box can be filled in n ways (because any one of the n items can be used to fill this box). Having filled the first box, to fill the second box we now have only $(n-1)$ items; any one of these items can be used to fill the second box and hence the second box can be filled in $(n-1)$ ways;

similarly, the third box in $(n-2)$ ways and so on, the r^{th} box can be filled in $\{n-(r-1)\}$ ways, i.e., $[n-r+1]$ ways. Hence, from the Fundamental Rule, all the r boxes together can be filled up in $n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-r+1)$ ways

$$\text{So, } {}^nP_r = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-r+1)$$

This can be simplified by multiplying and dividing the right hand side by $(n-r)(n-r-1) \dots 3 \cdot 2 \cdot 1$ giving us

$$\begin{aligned} {}^nP_r &= n(n-1)(n-2) \dots [n-(r-1)] \\ &= \frac{(n-1)(n-2) \dots [n-(r-1)] \cdot (n-r) \dots 3 \cdot 2 \cdot 1}{(n-r) \dots 3 \cdot 2 \cdot 1} = \frac{n!}{(n-r)!} \end{aligned}$$

The number of permutations of n distinct items taking r items at a time is

$${}^nP_r = \frac{n!}{(n-r)!}$$

If we take n items at a time, then we get nP_n . From a discussion similar to that we had for filling the r boxed above, we can find that nP_n is equal to $n!$.

The first box can be filled in n ways, the second one is $(n-1)$ ways, the

third one is $(n-2)$ ways and so on, then the n^{th} box in 1 way; hence, all the n boxes can be filled in

$n(n-1)(n-2).....3.2.1$ ways i.e., $n!$ ways.

Hence,

$${}^n P_n = n!$$

But if we substitute $r = n$ in the formula for ${}^n P_r$, then we get ${}^n P_n = \frac{n!}{0!}$;

since we already found that ${}^n P_n = n!$, we can conclude that $0! = 1$

6.4 Number of Combinations of n Dissimilar things Taken r at a Time

Let the number of combinations ${}^n C_r$ be x . Consider one of these x combinations. Since this is a combination, the order of the r items is not important. If we now impose the condition that order is required for these r items, we can get $r!$ arrangements from this one combination. So each combination can give rise to $r!$ permutations. x combinations will thus give rise to $x \cdot r!$ permutations. But since these are all permutations of n things taken r at a time, this must be equal to ${}^n P_r$.

$$\text{So, } {}^n P_r = \frac{n!}{(n-r)!}$$

$$\text{Hence, } {}^n C_r = \frac{n!}{r!(n-r)!}$$

The number of combinations of n dissimilar things taken all at a time is 1.

Out of n things lying on a table, if we select r things and remove them from the table, we are left with $(n-r)$ things on the table – that is, whenever r things are selected out of n things, we automatically have another selection of the $(n-r)$ things. Hence, the number of ways of making combinations taking r out of n things is the same as selecting $(n-r)$ things out of n given things, i.e.,

$${}^n C_r = {}^n C_{n-r}$$

When we looked at ${}^n P_r$, we imposed two constraints which we will now release one by one to find out the number of permutations.

6.5 Number of Arrangements of n Items of Which p are of one Type, q are of the Second Type and the Rest are Distinct

The number of ways in which n things may be arranged taking them all at a time, when p of the things are exactly alike and of one kind, q of them

exactly alike and of another kind, r of them exactly alike and of a third kind, and the rest all distinct is $\frac{n!}{p!q!r!}$

6.6 Total Number of Combinations

Out of n given items, the number of ways of selecting one or more items is where we can select 1 or 2 or 3 and so on n items at a time; hence the number of ways is ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$

This is called "the total number of combinations" and is equal to $2^n - 1$, where n is the number of items.

Number of ways of selecting one or more items from n given items is $2^n - 1$

- (i) Dividing given items into groups
- (ii) Dividing $(p + q)$ items into two groups of p and q items respectively.
- (iii) Out of $(p + q)$ items, if we select p items (which can be done in ${}^{p+q}C_p$ ways), then we will be left with q items, i.e., we have two groups of p and q items respectively. So, the number of ways of dividing $(p + q)$ items into two groups of p and q items respectively is equal to ${}^{p+q}C_p$ which is equal to $\frac{(p+q)!}{p!q!}$
- (iv) The number of ways of dividing $(p + q)$ items into two groups of p and q items respectively is $\frac{(p+q)!}{p!q!}$
- (v) The number of ways of dividing $2p$ items into two equal groups of p each of $\frac{(2p)!}{(p!)^2}$ where the two groups have distinct identities.
- (vi) The number of ways of dividing $2p$ items into two equal groups of p each is $\frac{(2p)!}{2!(p!)^2}$ where the two groups do not have distinct identity.
- (vii) The number of ways in which $(p + q + r)$ things can be divided into three groups containing p, q and r things respectively is $\frac{(p+q+r)!}{p!q!r!}$

If $p = q = r$, i.e., if we have to divide the given items into three EQUAL groups, then we have two cases where the three groups are distinct and where the groups are not distinct.

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When the three groups are distinct, the number of ways is $\frac{(3p)!}{(p!)^3}$

When the three groups are not distinct, then the number of ways is $\frac{(3p)!}{3!(p!)^3}$

6.7 Circular Permutations

When n distinct things are arranged in a straight line taking all the n items, we get $n!$ permutations. However, if these n items are arranged in a circular manner, then the number of arrangements will not be $n!$ but it will be less than that. This is because in a straight line pattern, if we have an arrangement ABCDE, and if we move every item one place to the right (in cycle order), the new arrangement that we get, EABCD, will not be the same as ABCDE and this also is counted in the $n!$ permutations that we talked about. However, if we have an arrangement ABCDE in a circular fashion, by direction, we still get the same arrangement ABCDE. So, if we now take $n!$ as the number of permutations, we will be counting the same arrangement more than once.

The number of arrangements in circular fashion can be found out by first fixing the position of one item. Then the remaining $(n-1)$ items can be arranged in $(n-1)!$ ways. Now even if we move these $(n-1)$ items by one place in the clockwise direction, then the arrangement that we get will not be the same as the initial arrangement because one item is fixed and it does not move.

Hence, the number of ways in which n distinct things can be arranged in a circular arrangement in $(n-1)!$

If we take the case of five people, A, B, C, D and E, sitting around a table, then the two arrangements ABCDE (in clockwise direction) and AEDCB (the same order but in anticlockwise direction) will be different and distinct. What is the difference between different and distinct? Here we say that the clockwise and anticlockwise arrangements are different. However, if we consider the circular arrangement of a necklace made of five precious stones A, B, C, D and E, the two arrangements talked of above will be the same because we take one arrangement and turn the necklace around (front to back), then we get the other arrangement. Here, we say that there is no difference between the clockwise and anticlockwise arrangements. In this case the number of arrangements will be half of what it is in the case where the clockwise and anticlockwise arrangements are different.

The number of circular arrangement of n distinct item is $(n-1)!$ if there is a DIFFERENCE between the clockwise and anticlockwise arrangements, and $\frac{(n-1)!}{2}$ if there is NO DIFFERENCE between the clockwise and

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anticlockwise.

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6.8 Sum of all Numbers Formed From the Given Digits

- (i) If n distinct digits are used to make all the possible n – digit numbers, we get $n!$ numbers. We now want to find out the sum, of these $n!$ numbers. Let us take an example and understand how this can be done and then look at it as formula.
- (ii) To find the sum of all the four-digit numbers formed using the digits 2, 3, 4 and 5 without repetition.
- (iii) We can form a total of $4!$ or 24 numbers. When we add all these numbers, let us look at the contribution of the digit 2 to the sum.
- (iv) When 2 occurs in the thousands place in a particular numbers, its contribution to the total will be 2000. The number of numbers that can be formed with 2 in the thousands place is $3!$, i.e. 6 numbers. Hence, when 2 is in the thousands place, its contribution to the sum is $3! \times 2000$
- (v) Similarly, when 2 occurs in the hundreds place in a particular number, its contribution to the total will be 200 and since there are $3!$ numbers with 2 in the hundreds place, the contribution 2 makes to the sum when it comes in the hundreds place is $3! \times 200$.
- (vi) Similarly, when 2 occurs in the tens and units place respectively, its contribution to the sum is $3! \times 20$ and $3! \times 2$ respectively. Thus the total contribution of 2 to the sum is $3! \times 2000 + 3! \times 200 + 3! \times 20 + 3! \times 2$, i.e., $3! \times 2222$. This takes care of the digit 2 completely.
- (vii) In a similar manner, the contribution of 3, 4 and 5 to the sum will be $3! \times 3333$, $3! \times 4444$ and $3! \times 5555$ respectively.
- (viii) The sum can now be obtained by adding the contributions of these four digits. Hence, the sum of the numbers formed by using the four digits is $3! \times (2222 + 3333 + 4444 + 5555)$, i.e.,

$$3! \times (2 + 3 + 4 + 5) \times 1111$$

We can now generalize the above as

If all the possible n – digit numbers using n distinct digits are formed, the sum of all the numbers so formed is equal to $(n-1)! \times \{\text{sum of the } n \text{ digits}\} \times \{1111 \dots n \text{ times}\}$.

6.9 Rank of Word

Finding the rank of a given word is basically finding out the position of the word when all possible words have been formed using all the letters of this

word exactly once and arranged in alphabetical order as in the case of a dictionary. Let us understand this by taking an example.

Let us look at the word "POINT". The letters involved here, when taken in alphabetical order are I, N, O, P, T.

To arrive at the word "POINT", initially we have to go through the words that begin with I, then all those that begin with N and those that begin with O, which are 4! In each case then we arrive at words that begin with PI, PN which are 3! in each case. Then we arrive at the word POINT.

There are $3 \times 4! + 2 \times 3! = 84$ words that precede the word POINT i.e., POINT is the 85th word. Hence the rank of the 'POINT' is 85.

6.10 The Number of Diagonals in an n - sided Polygon

An n - sided regular polygon has n vertices. Joining any two vertices we get a line of the polygon, which are nC_2 in number. Of these nC_2 lines, n of them are sides. Hence the diagonals are ${}^nC_2 - n = \frac{n(n-3)}{2}$

6.11 Number of Internal Solution of the Equation $x_1 + x_2 + \dots + x_n = s$

Consider the equation

$$x_1 + x_2 + x_3 = 10$$

- (i) If we consider all possible integral solutions of the equation, there are infinitely many. But the number of positive (or non-negative) integral solutions is finite.
- (ii) We would like the number of positive integral solutions of this equation, i.e., values of (x_1, x_2, x_3) such that each $x_i > 0$
- (iii) We imagine 10 identical objects arranged on a line. There are 9 gaps between these 10 objects. If we choose any two of these gaps, we are effectively splitting the 10 identical objects into 3 parts of distinct identity. Conversely, every split of these 10 objects corresponds to selection of 2 gaps out of the 9 gaps.
- (iv) Therefore, the number of positive integral solutions is 9C_2 . In general, if $x_1 + x_2 + \dots + x_n = s$ where $s \geq n$, the number of positive integral solutions is ${}^{s-1}C_{n-1}$.
- (v) For $x_1 + x_2 + x_3 + \dots + x_n = s$ where $s \geq 0$, the number of positive integral solutions (when $s \geq n$) is ${}^{s-1}C_{n-1}$ and the number of non-negative integral solutions is ${}^{n+s-1}C_{n-1}$.

Miscellaneous

- (i) The total number of ways in which a selection can be made by taking some or all out of $p + q + r + \dots$ things where p are alike of one kind, r alike of a third kind and so on it
- $$\left[\{(p+q)(q+1)(r+1)\dots\} - 1 \right].$$

(ii) ${}^{n+1}C_r = {}^nC_r + {}^nC_{r-1}$ and ${}^nP_r = r \cdot {}^{n-1}P_{r-1} + {}^{n-1}P_r$

Examples:

- (a) How many four letter words can be formed using the letters of the word "ROAMING"?

Exp: None of the letters in the word are repeated.

$\therefore$ The number of four letter words that can be formed

$$= {}^7P_4 = \frac{7!}{3!} = (7)(6)(5)(4) = 840.$$

- (b) How many three letter words can be formed using the letters of the word "PRACTICES"?

Exp: PRACTICES. The contributions, the number of combinations and the number of permutations (for each combination) as well as the total are tabulated below.

Combination	Number of combination	Number of Permutations for each combination	Total number of permutations
x, y, z	${}^8C_3 = 56$	6	336
x, x, y <i>i.e., c, c, y</i>	${}^7C_1 = 7$	3	21
			357

- (c) In a party, each person shook hands with every other person present. The total number of hand shakes was 28. Find the number of people present in the party.

Exp: Let the number of people present in the party be n

Method-1: The first person shakes hands with a total of $(n-1)$ persons, the second with $(n-2)$ other people and so on.

The total number of hand shakes is

$$(n-1) + (n-2) + \dots + 2 + 1$$

$$\frac{n(n-1)}{2} = 28 \text{ (given)} \Rightarrow n = 8.$$

Method-2: Number of hands shakes = Number of ways of selecting 2 people out of $n = {}^nC_2$

$${}^nC_2 = 28$$

$$\frac{n(n-1)}{2!} = 28$$

$$\Rightarrow n = 8$$

Direction for Example Q. No. 4 to Q. No. 8: The following examples are based on the data below.

The letters of NESTLE are permuted in all possible ways.

- (d) How many of these words begin with T?

Exp: NESTLE has 6 letters of which the letter E occurs two times. Therefore the required number of words = Number of ways of filling N,

$$\text{E, S, E and L in the second to sixth positions} = \frac{5!}{2!} = 60.$$

- (e) How many of these words begin and end with E?

Exp: The required number of words = The number of ways of filling N, S, T and L in the second to fifth positions = $4! = 24$

- (f) How many of these words begin with S and end with L?

Exp: The required number of words = The number of ways of filling N, E, T and E in the second to fifth positions = $\frac{4!}{2!} = 12.$

- (g) How many of these words neither begin with S nor end with L?

Exp: The required number of words = The total number of words which can be formed using the letters N, E, S, T and E – (Number of words which begin with S or end with L)

$$= \frac{6!}{2!} - (\text{Number of words beginning with S} + \text{Number of words ending with L} - \text{Number of words beginning with S and ending with L})$$

$$= \frac{6!}{2!} - \left(\frac{5!}{2!} + \frac{5!}{2!} - \frac{4!}{2!} \right)$$

$$= 360 - (60 + 60 - 12) = 252.$$

- (h) How many of these words begin with T and do not end with N?

Exp: The required number of words = The number of words beginning with T - The number of words beginning with T and ending with N

$$= \frac{5!}{2!} - \frac{4!}{2!} = 48.$$

Directions for example Q. No. 8–11: The following examples are based on the data below.

Raju wrote 7 letters A, B, C, D, E, F and G on a black board.

- (i) How many 4-letter words can be formed using these letters such that at least one letter of the word is a vowel?

Exp: The vowels, the number of ways in which the consonants can be selected and the number of words are tolerated letters.

Vowels	No. of ways of selecting consonant	Number of words
A	${}^5C_3 = 10$	$10(24)$
E	${}^5C_3 = 10$	$10(24)$
AE	${}^5C_2 = 10$	$10(24)$
		$10(72) = 720$

- (j) How many 7 letter words can be formed using these letters such that the letters at the words end are adjacent consonants?

Exp: B and C, C and D as well as F and G are adjacent consonants. The letters in each of these pairs can be arranged in $2! = 2$ ways. The remaining letters can be arranged in $5! = 120$ ways.

$\therefore$ The required number of words = $(3)(2)(120) = 720$.

- (k) How many 7 letters words can be formed using these letters such that the letter at one of its ends is a vowel and that at the other end is a consonant?

Exp: If the first letter is a vowel, it has 2 possibilities.

In this case, the last letter has 5 possibilities.

The remaining 5 letters can be arranged in $5! = 120$ ways.

$\therefore$ The total number of words = $(2)(5)(120)$.

If the last letter is a vowel, it similarly follows that the number of words = $(2)(5)$ and (120) .

So, the total number of words = $(2)(2)(5)(120) = 2400$.

Direction for example Q. No. 12-13: The following examples are based on the data below.

All possible four-digit numbers are formed using the digits 1, 2, 3 and 4 without repeating any digit.

- (l) How many of these numbers have the even digits in even places?

Exp: There are two even places.

$\therefore$ The even digits can be arranged in $2! = 2$ ways.

∴ The required number of words = $(2)(2) = 4$.

- (m) If all the numbers are arranged in an ascending order of magnitude, find the position of the number 3241.

Exp: If the first digit is 1 or 2, in each case, the remaining digits can be arranged in $3!$ or 6 ways.

∴ a total of $(2)(6) = 12$ numbers have their first digit as 1 or 2.

If the first digit is 3, the possible numbers in ascending order are 3124, 3142, 3214, 3241.

∴ The position of 3241 = 16

Direction for examples Q. No. 14-17: The following examples are based on the data below.

A committee of 5 is to be formed from 4 women and 6 men.

- (n) In how many ways can it be formed if it consists of exactly 2 women?

Exp: The committee must have 2 women and 3 men.

∴ The required number of ways = ${}^4C_2 \cdot {}^6C_3 = 120$

- (o) In how many ways can it be formed if it consists of more women than men?

Exp: The committee must have either 4 women and 1 man or 3 women and 2 men.

∴ The required number of ways

$$= {}^4C_2 \cdot {}^6C_1 + {}^4C_3 \cdot {}^6C_2 = 6 + 60 = 66.$$

- (p) Find the number of four-digit numbers, which can be formed using four of the digits 0, 1, 2, 3 and 4 without repetition.

Exp: The first digit has 4 possibilities (1, 2, 3 and 4).

The second digit has 4 possibilities (0 and any of the three digits not used as the first digit).

The third digit has 3 possibilities.

The last digit has 2 possibilities.

∴ The required number of numbers = $(4)(4)3(2) = 96$.

- (q) The number of diagonals of a regular polygon is four times the number of its sides. How many sides does it have?

Exp: Let the number of sides in the polygon be n .

$$\frac{n(n-3)}{2} = 4n$$

$$n(n-11) = 0$$

$$n > 0$$

$$\therefore n - 11 = 0$$

$$\therefore n = 11$$

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Concept Review Questions

1. If ${}^nC_2 = {}^nC_{10}$ then the value of n is

- (a.) 10
(b.) 2
(c.) 8
(d.) 12

Ans: d

2. The relation between nP_r and nC_r is

- (a.) ${}^nP_r = {}^nC_r$
(b.) $r \cdot {}^nP_r = {}^nC_r$
(c.) ${}^nP_r = r! \cdot {}^nC_r$
(d.) ${}^nP_r \times r! = {}^nC_r$

Ans: c

3. The number of 7-letter words that begin with R when each letter occurs only once is

- (a.) $6(6!)$
(b.) $7! \cdot 2!$
(c.) $6!$
(d.) $2(7!)$

Ans: c

4. Find the number of ways of selecting a team of 5 people from a group of 8.

- (a.) 8C_3
(b.) 8P_5
(c.) $8!$
(d.) $5!$

Ans: a

5. Find the number of ways of studding 10 beads to form a necklace.

- (a.) $\frac{9!}{2!}$
(b.) $9!$
(c.) $9! \cdot 2!$

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(d.) $\frac{10!}{2}$

Ans: a

6. The number of rectangles that can be formed on a 8×8 chessboard is

(a.) 2194

(b.) 1284

(c.) 1196

(d.) 1296

Ans: d

7. The number of squares that can be formed on a 8×8 chessboard is

(a.) 204

(b.) 220

(c.) 240

(d.) 210

Ans: a

8. Find the number of ways in which the letters of the word INCLUDE can be permuted so that no two vowels appear together.

(a.) $7! - 5!3!$

(b.) $7! - 4!2!$

(c.) $4! \times 3!$

(d.) $\frac{4! \times 5!}{2}$

(e.) $4! \times 5!$

Ans: d

9. In how many ways can 5 prizes be given away to 3 boys when each boy is eligible for one or more prizes?

(a.) 5^3

(b.) 3^5

(c.) 5P_3

(d.) 5C_3

(e.) 242

Ans: b

10. In how many ways can the letters of the word COMBINATION be permuted?

(a.) $11!$

(b.) $\frac{11!}{2!2!2!}$

(c.) $\frac{11!}{5!6!}$

(d.) $\frac{11!}{2!2!2!5!}$

(e.) $\frac{11!}{2!2!}$

11. Find the number of selections that can be made by taking 4 letters from the words INKLING.

(a.) 48

(b.) 38

(c.) 28

(d.) 18

(e.) 58

Ans: d

12. In how many ways can 6 boys and 6 girls sit around a circular table so that no two boys sit next to each other?

(a.) $(5!)^2$

(b.) $(6!)^2$

(c.) $5!6!$

(d.) $11!$

(e.) $12!$

Ans: c

13. Find the number of non-negative integers which satisfies the equation $x_1 + x_2 + x_3 + x_4 = 15$.

(a.) 216

(b.) 165

(c.) 364

(d.) 316

(e.) 818

Ans: e

14. The number of positive integral solutions to the equation $x + y + z = 20$ is

(a.) 131

(b.) 110

(c.) 55

(d.) 171

(e.) 141

Ans: d

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CHAPTER 7

PROBABILITY

7.1 Deterministic and Probabilistic Phenomena

- (i) If the result of an experiment is unique or certain, then the experiment can be said to be of a deterministic nature.
- (ii) If the result is not unique and can be any one of the several possible outcomes, such an experiment is called a random experiment. Random experiment is of probabilistic nature. For example, when we throw a die, we may get as the outcome any integer from 1 to 6. Hence, this can be termed as a random experiment.

7.2 Biased and Unbiased Experiments

- (i) When we throw a die, if there is any reason to believe that one of the numbers will turn up more frequently (or the likelihood of getting one or more numbers is more than that of the other numbers), we say it is a biased die. If all the six numbers are equally likely, then we call it an unbiased die. Similarly, a coin also can be termed as an unbiased or a biased coin depending on whether the head and tail are equally likely or not. We will be mostly looking at unbiased experiment. Hence, unless otherwise specified, in the discussion that follows, a coin or a die is to be treated as unbiased.
- (ii) The basic outcomes of an experiment are termed as occurrences. Hence, if we throw a die there are 6 occurrences. If we toss a coin there are 2 occurrences.
- (iii) Events are defined as a collection of one or more occurrences. For example, if we throw a die we can define the following 6 events:

Getting a 1

Getting a 2

Getting a 3

Getting a 4

Getting a 5

Getting a 6

Here, each of these events has one occurrence. Consider the following two events when a die is thrown:

- (a) Getting an even number
- (b) Getting an odd number

Each of these 2 events have 3 occurrences. So, in this case, we say that out of a total of occurrences in the experiment, 3 are favourable to the event of "getting an odd number" and 3 occurrences are favourable to the event of "getting an even number"

7.3 Equally Likely Events

Two events are said to be equally likely when there is no reason to expect any one rather than the other.

For example, when a die is thrown, any number from 1 to 6 may show up. In this trial, the six events are equally likely.

7.4 Compound Events

When two or more events are in relation with each other, they are known as Compound Events.

For example, when a die is thrown two times, the event of getting 3 in the first throw and 5 in the second throw is a compound event.

7.5 Definition of Probability

- (i) If in an experiment there are ' n ' occurrences out of which ' m ' occurrences are favourable to a particular event E , then the probability of the event E [written as $P(E)$] is defined as $P(E) = \frac{m}{n}$.
- (ii) For the event E , the non-occurrence of the event is denoted by $\bar{E}$. Out of a total of n occurrences, if m occurrences are favourable to the event E , it means that $(n - m)$ occurrences are not favourable to the event E , i.e. $(n - m)$ occurrences are favourable to the event $\bar{E}$.
- (iii) Hence, the probability of the event $\bar{E}$ will be $P(\bar{E}) = \frac{n - m}{n}$
(Complementary event).
- (iv) We can also see that $P(\bar{E}) + P(E) = 1$. In some problems when we are asked to find out the probability of an event $\bar{E}$, it becomes easier to calculate the probability of E and subtract that probability from 1 to get $P(\bar{E})$. (Which we will see in some of the examples).
- (a) Probability $[P(E)]$ of the happening of an event E is known as the probability of success and the probability $[P(\bar{E})]$ of the non-happening of the event is the probability of failure.

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(b) If $P(E) = 1$, the event is called a certain event and if $P(E) = 0$ the event is called an impossible event.

(c) Instead of saying that the chance of happening of an event is $\frac{m}{n}$, we can also say that the odds in favour of the event are m to $(n - m)$. Similarly, if $\frac{m}{n}$ is the probability of the occurrence of an event, since $\frac{(n - m)}{n}$ is the probability of non-occurrence of the event, we can also say that the odds against the event are $(n - m)$ to m .

Set-1: Getting a 1; getting a 2; getting a 3; getting a 4; getting a 5; getting a 6.

Set-2: Getting an even number; getting an odd number.

Set-3: Getting a number > 3 , getting a number < 3 .

Set 4: Getting a number ≥ 3 , getting a number ≤ 3 .

(i) If we compare the 1st and 3rd set of events, one main difference is that the events of set 1 include all possible occurrences of the experiments without leaving any occurrence, whereas, the events of set 3 do not cover all the possible occurrences of the experiment (because the occurrence of getting a 3 is not covered in either one of the two events in set 3).

(ii) Events that together cover all the possible occurrences of an experiment are termed as COLLECTIVELY EXHAUSTIVE events. The events of set 2 and 4 are also collectively exhaustive events.

(iii) If we compare the events of a set 1 and set 4, the main difference is that in the events of set 1 no occurrence appears in more than one event, whereas, in the events of set 4 some occurrences appear in more than one event (in this case, the occurrence 3 appears in both the events of set 4).

(iv) If the occurrence of one event precludes the possibility of the occurrence of any other event, such events are called MUTUALLY EXCLUSIVE events.

(v) So, in a set of mutually exclusive events, any occurrence of the experiment cannot come in more than one event.

(vi) As is clear, set 2 and 3 are also mutually exclusive events.

(vii) So, set 1 and 2 are both mutually exclusive and collectively exhaustive.

(viii) If we have a set of events which are both mutually exclusive and collectively exhaustive, then the sum of the probabilities of such events will be equal to 1.

(ix) The two events we discussed above are mutually exclusive and

collectively exhaustive as per the definitions we just looked at.

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7.6 Addition Theorem on Probability

- (i) If there are two sets A and B , we know that the number of elements in $A \cup B$ is given by $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- (ii) A similar relationship exists in Probability Theorem (called the Addition Theorem of Probability) as $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, where A and B are two events.
- (iii) If A and B are mutually exclusive events (i.e., A and B are disjoint sets), then $P(A \cap B) = 0$
- (iv) If there are two mutually exclusive events A and B , then $P(A \cup B) = P(A) + P(B)$

7.7 Independent Events

- (i) Two events E_1 and E_2 are said to be independent, if the occurrence of the event E_2 is not affected by the occurrence or non-occurrence of the event E_1
- (ii) If A and B are two independent events, then $P(A \cap B) = P(A) \cdot P(B)$

Examples:

- (a) Two dice are rolled simultaneously. Find the probability that one of them shows a number which is at least 4 and the other shows a number which is at most 3.

Exp: When two dice are thrown simultaneously, total number of outcomes = $(6)(6) = 36$.

Out of these, favourable cases are $(4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (5, 3), (6, 1), (6, 2)$ and $(6, 3)$. i.e., 9

$$\therefore \text{Required probability} = 9 \left(\frac{1}{6} \right) \left(\frac{1}{6} \right) \times 2 = \frac{1}{2}$$

- (b) Two dice are rolled simultaneously. Find the probability that the sum of the numbers on them is not 8.

Exp: Probability of sum of the numbers is not 8 = $1 - \text{Probability of sum of the numbers on them is 8}$

= favourable cases for sum of the number is 8 are $(2, 6), (3, 5), (4, 4), (5, 3)$ or $(6, 2)$

$$\text{Probability (sum of the numbers on them is 8)} = 5 \left(\frac{1}{6} \right) \left(\frac{1}{6} \right) = \frac{5}{36}$$

$$\text{Required probability} = 1 - \frac{5}{36} = \frac{31}{36}$$

- (c) Three fair dice are rolled simultaneously. Find the probability that the first die shows a composite number, second die shows an odd number and third die shown an odd prime number.

Exp: Required probability = Probability of first die shows a composite number $\times$ probability of second die shows an odd number $\times$ probability

$$\text{of third die shows an odd prime number} = \frac{2}{6} \left(\frac{3}{6} \right) \left(\frac{2}{6} \right) = \frac{1}{18}$$

- (d) Four dice are rolled simultaneously. Find the probability that the sum of the numbers exceeds 22.

$$\text{Exp: Total number of cases} = 6^4 = 1296$$

The various combinations for the sum exceeding 22 and the corresponding number of arrangements in each case are as follows:

Sum being 23 – (6, 6, 6, 5), (6, 6, 5, 6), (6, 5, 6, 6) or (5, 6, 6, 6)

Sum being 24 – (6, 6, 6, 6)

$$\text{Required probability} = 5 \left(\frac{1}{6} \right)^4 = \frac{5}{1296}$$

- (e) Five fair coins are tossed together. Find the probability of getting exactly 4 heads.

Exp: The event of getting exactly 4 heads will be the combination of 4 heads and 1 tail. Number of arrangements possible with this combination

$${}^5C_4 = \frac{5!}{4!1!} = 5$$

$$\therefore \text{Required probability} = \frac{5}{2^5} = \frac{5}{32}$$

- (f) Five fair coins are tossed together. Find the probability of getting at most 3 heads.

Exp: Probability of getting at most 3 heads

$$= 1 - \text{probability of getting at least 4 heads}$$

$$= 1 - [(\text{probability of getting 4 heads}) + (\text{probability of getting 5 heads})]$$

$$= 1 - \left[{}^5C_4 \left(\frac{1}{32} \right) + \frac{1}{32} \right]$$

$$= 1 - \frac{6}{32} = \frac{13}{16}$$

(g) A card is drawn from a well shuffled pack of cards. Find the probability of it being

- (i) a jack
- (ii) a black numbered card
- (iii) a diamond
- (iv) a red king

Exp: Number of ways of selecting one card out of 52 cards = ${}^{52}C_1 = 52$

(i) One jack can be drawn from 4 jacks in ${}^4C_1 = 4$ ways

∴ The required probability

$$= \frac{{}^4C_1}{{}^{52}C_1} = \frac{4}{52} = \frac{1}{13}$$

(ii) A pack of cards has 26 black cards of which 18 are numbered. A black numbered card can be drawn in ${}^{18}C_1 = 18$ ways.

∴ The required probability

$$= \frac{{}^{18}C_1}{{}^{52}C_1} = \frac{18}{52} = \frac{9}{26}$$

(iii) A pack of cards has 13 diamonds.

A diamond can be drawn in ${}^{13}C_1 = 13$ ways.

∴ The required probability

$$= \frac{{}^{13}C_1}{{}^{52}C_1} = \frac{13}{52} = \frac{1}{4}$$

(iv) A pack of cards had 2 red kings.

A red king can be drawn in ${}^2C_1 = 2$ ways.

$$= \frac{{}^2C_1}{{}^{52}C_1} = \frac{2}{52} = \frac{1}{26}$$

(h) Two cards are drawn simultaneously from a well shuffled pack of cards. Find the probability of both being.

- (i) Kings
- (ii) Honours
- (iii) Black honours

Exp: Two cards can be drawn from a pack of cards in ${}^{52}C_2$ ways.

- There are 4 kings in a pack of cards. Two kings can be drawn in 4C_2 ways.

- The required probability = $\frac{{}^4C_2}{{}^{52}C_2}$

- There are 16 honours in a pack of cards.

- The required probability = $\frac{{}^{16}C_2}{{}^{52}C_2}$

- There are 8 black honours in a pack of cards

- The required probability = $\frac{{}^8C_2}{{}^{52}C_2}$

- (i) Two cards are drawn at random from a pack of cards. Find the probability that one of them is a numbered card and the other is a queen.

Exp: Two cards can be drawn from a pack of cards in ${}^{52}C_2$ ways.

A numbered card can be drawn in ${}^{36}C_1$ ways.

And a queen card can be drawn in 4C_1 ways.

The required probability = $\frac{{}^{36}C_1 {}^4C_1}{{}^{52}C_2}$

- (j) Two cards are drawn simultaneously from a pack of cards. Find the probability that both are hearts or both are diamonds.

Exp: Both cards should be hearts or diamonds.

These are mutually exclusive events.

Let X denote the event of getting both hearts and Y denote the event of getting both diamonds.

$$P(X) = \frac{{}^{13}C_2}{{}^{52}C_2} \text{ and } P(Y) = \frac{{}^{13}C_2}{{}^{52}C_2}$$

$$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$$

$$2 \left(\frac{{}^{13}C_2}{{}^{52}C_2} \right) \left[\because P(X \cap Y) = 0 \right]$$

- (k) Three cards are drawn simultaneously from a pack of cards. Find the probability that one of them is red, another is a spade and the third is a club.

Exp: There are 26 red cards, 13 spades and 13 clubs in a pack of cards.

$$\therefore \text{The required probability} = \frac{{}^{26}C_1 {}^{13}C_1 {}^{13}C_1}{{}^{52}C_3}$$

(l) Two balls are drawn in succession without replacement from a box containing 3 black balls and 6 blue balls. Find the probability that

- (i) The first ball is black and the second ball is blue.
- (ii) Both are blue.

Exp: The first ball can be drawn in ${}^9C_1 = 9$ ways and as it is not replaced, the second ball can be drawn in ${}^8C_1 = 8$ ways.

- A black ball can be drawn in ${}^3C_1 = 3$ ways in the first draw. A blue ball can be drawn in ${}^6C_1 = 6$ ways in the second draw.

$$\therefore \text{The required probability} = \left(\frac{3}{9}\right)\left(\frac{6}{8}\right) = \frac{1}{4}$$

- A blue ball can be drawn in ${}^6C_1 = 6$ ways in the first draw.

Since this ball is not replaced, a blue ball can be drawn in ${}^5C_1 = 5$ ways in the second draw.

$$\therefore \text{The required probability} = \left(\frac{6}{9}\right)\left(\frac{5}{8}\right) = \frac{5}{12}$$

(m) A bag contains 3 blue balls, 4 green balls and 5 red balls. A ball is drawn at random. Find the probability that it is

- (a) Not a green ball
- (b) A red ball

Exp: A ball can be drawn from 12 balls in ${}^{12}C_1 = 12$ ways.

- (i) A ball other than a green ball ($3 + 5 = 8$) can be drawn in 8C_1 ways. Hence, the probability of not drawing a green ball

$$= \frac{{}^8C_1}{{}^{12}C_1} = \frac{2}{3}$$

- (ii) A red ball out of the 5 red balls present in the bag can be drawn in ${}^5C_1 = 5$ ways.

$$\text{Hence, the probability of drawing a red ball} = \frac{{}^5C_1}{{}^{12}C_1} = \frac{5}{12}$$

(n) A bag contains 3 blue balls, 4 green balls and 5 red balls. Three balls are drawn simultaneously at random. Find the probability that

- (a) The balls are of different colour
- (b) Two are blue and one is green.
- (c) The balls are of the same colour.

Exp: Three balls can be drawn from 12 balls in ${}^{12}C_3$ ways.

- (i) We can draw a blue ball out of 3 blue balls in 3C_1 ways, a green ball out of 4 green balls in 4C_1 ways and a red ball out of 5 red balls in 5C_1 ways.

$$\text{The required probability} = \frac{{}^3C_1 {}^4C_1 {}^5C_1}{{}^{12}C_3}$$

- (ii) Two blue balls and one green ball can be drawn in ${}^3C_2 \cdot {}^4C_1$ ways.

$$\therefore \text{The required probability} = \frac{{}^3C_2 {}^4C_1}{{}^{12}C_3} = \frac{3}{55}$$

- (iii) 3 blue balls or 3 green balls or 3 red balls can be drawn in 3C_3 or

$${}^4C_3 \text{ or } {}^5C_3 \text{ ways required probability} = \frac{{}^3C_3 + {}^4C_3 + {}^5C_3}{{}^{12}C_3} = \frac{3}{44}$$

- (o) Mohan had 5 fifty rupee notes, 3 hundred rupee notes and 4 five hundred rupee notes in his pocket. Two notes were drawn at random. Find the odds in favour of both being fifty rupee notes.

Exp: Total number of ways of drawing two notes from a pocket containing 12 notes in ${}^{12}C_2$ ways. The number of ways in which two fifty rupee notes can be drawn = 5C_2

$$\text{The probability of choosing two fifty rupee notes} = \frac{{}^5C_2}{{}^{12}C_2} = \frac{5}{33}$$

$\therefore$ Odds in favour of the event = favorable ways: unfavorable ways = 5:28

- (p) There are 4 red, 6 blue and 8 white balls in a bag. Raju drew 2 balls from it one after another. Find the probability of the second ball drawn being blue if the first ball is

- (i) Replaced
(ii) Not replaced.

Exp (i): the required probability = Probability of drawing any ball from the bag as the first ball). (Probability of drawing a blue ball from it as

$$\text{the second ball} = 1 \left(\frac{6}{18} \right) = \frac{1}{3}$$

- (ii) There are two cases. The first ball is blue or it is not blue.

$\therefore$ The probability that the second ball is blue is

$$\frac{6}{18} \left(\frac{5}{17} \right) + \frac{12}{18} \left(\frac{6}{17} \right) = \frac{6}{18} \left(\frac{5}{17} + \frac{12}{17} \right) = \frac{6}{18} = \frac{1}{3}$$

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Probability

CHAPTER 8

INTRODUCTION TO DATA INTERPRETATION

8.1 Data Table

Here data is presented in the form of simple table. While any type of data can be presented in tabular form, that too in a very accurate manner, interpreting the data in tabular form is more difficult and time consuming than the other modes, all of which are basically pictorial or graphical in presentation.

Data tables can be of a number of types. They can be of a single table variety or combination of tables. Some examples of tables are given below.

Railway Time Table- Coromandel Express			
Place	Cumulative mileage	Arrival Time (in hrs.)	Departure Time (in hrs.)
Madras	0	-	08.00
Nellore	200	11.20	11.30
Vijayawada	525	15.30	16.00
Rajamundry	700	19.20	19.30
Visakhapatnam	1100	01.10	01.30
Bhubaneshwar	1450	03.45	04.00
Kharagpur	1600	07.25	07.30
Kolkata	1925	09.30	-

From the above Time Table, we can obtain the following:

- Distance between various stations.
- Total idle time as a proportional of total travel time.
- Average speed between stations as well as over the entire journey.
- Minimum and maximum speeds between two stations.

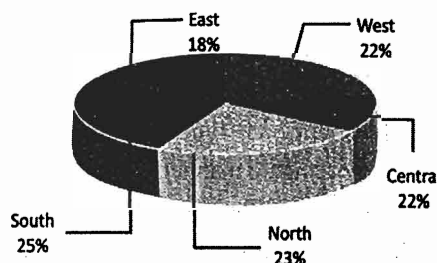
8.2 Pie-Chart

This is probably the simplex of all pictorial forms of data presentation. Hence, total quantity to be shown is distributed over one complete circle or 360 degrees. In pie-charts, data is essentially presented with respect to only

one parameter (unlike in two and 3-dimensional graphs, described later). This form essentially presents shares of various elements as proportion or percentage of the total quantity. Each element or group in the pie-chart is represented in terms of quality (or value, as the case may be) or as the angle made by the sector representing the elements or as a proportion of the total or as a percentage of the total.

Chart-1 gives distribution of the population in different geographical zones.

Distribution of Population in Geographical Zones



From the above pie chart, we can calculate the following:

- Population in any zone given the total population.
- Population of any zone as a percentage of that of another zone.
- Percentage increase in the total population given the percentage increase in the population of one or more zones.

Pie charts are also very frequently used in combination with other forms of data or along with the Pie-Charts.

8.3 Two-Dimensional Graphs

This is essentially used for continuous data but can also be used for depicting discrete data provided we understand the limitation. Also, known as Cartesian Graphs, they represent variation of one parameter with respect to another parameter each shown on a different axis. These types of graphs are useful in studying the rate of change or understanding the trends through extrapolations.

Motion Graph

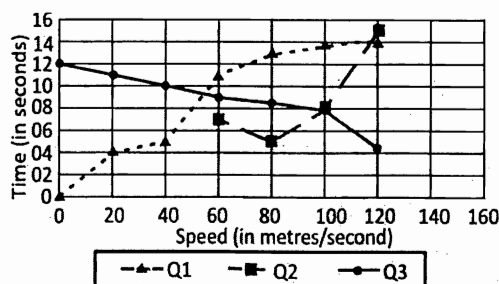


Chart represents another type of two-dimensional graph which is mostly used to depict scientific data like speed, velocity, vector etc. In the graph speed trends of three bodies Q 1, Q 2, Q 3 is given along with their actual

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path of motion.

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8.4 Bar-Charts

This is a type of graph used mostly to depict data in a discrete way. They are accurate and comparison of variables is very convenient.

Refrigerator Sales of Company ABC

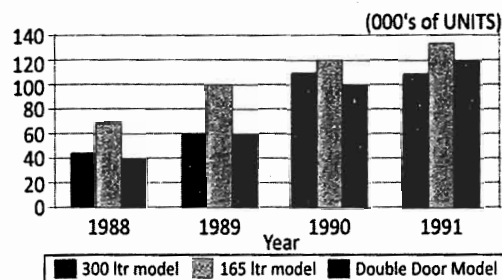


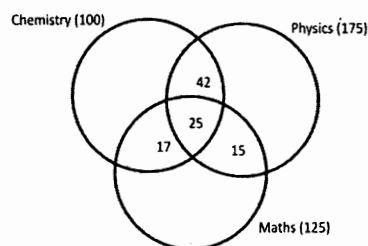
Chart 5 shows model wise sales of refrigerators during four years. From this graph we can obtain the following:

- Percentage contribution of each model to the company's total sales for four years.
- Relative increase or decrease in the share of each model.
- Sales trend of various models.

8.5 Venn-Diagrams

You must be familiar with the concept of sets. Data is represented in the form of Venn-Diagrams when operations have to be carried out on different distinct sets of elements each following a different functional rule. All the elements in a set follow the same functional rule. By Set Union and Intersection operations, you can establish new sets from the existing sets.

For example: consider three of the courses Physics, Chemistry and Maths offered to B.Sc. students of various groups



From the chart, you can arrive at the number of students who are studying only one out of the above three subjects.

Example:

These questions are based on the following table, which gives the details of the sports which students in all the classes of a school like.

The table gives the number of students in each class and the percentage of students in it who like Cricket, Volleyball, Basketball and Football.

Class	Number of students	Cricket	Volleyball	Basket ball	Foot ball
6	120	60%	70%	50%	60%
7	140	50%	60%	60%	50%
8	160	40%	65%	55%	45%
9	180	65%	75%	65%	55%
10	240	70%	80%	75%	45%

1. How many students in the school like cricket?

- (a.) 436
- (b.) 432
- (c.) 491
- (d.) 511
- (e.) 540

Ans: c

Exp: Number of students who like Cricket

$$\begin{aligned}
 &= \frac{60}{100}(120) + \frac{50}{100}(140) + \frac{40}{100}(160) \\
 &+ \frac{65}{100}(180) + \frac{70}{100}(240) \\
 &= 491
 \end{aligned}$$

2. By what percentage is the number of students who like Volleyball in class 6 more/less than those who like Basketball in class 10?

- (a.) 40%
- (b.) 50%
- (c.) 53.33%
- (d.) 56.67%
- (e.) None of these

Ans: c

Exp: Number of students who like Volleyball in class 6

$$= \frac{70}{100}(120) = 84$$

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Number of students who like Basketball in class 10 = $\frac{75}{100}(240) = 180$

84 is less than 180 by $\frac{180-84}{180}(100)$

= 53.33%.

3. The number of students who like cricket in class 7 is what percentage of the number of students who like Football in class 8?

- (a.) 88%
(b.) 93.5%
(c.) 95.6%
(d.) 97.2%
(e.) 99.2%

Ans: d

Exp: Number of student who like Cricket in class 7 = $\frac{50}{100}(140) = 70$

Number of students who like Football in class 8 = $\frac{45}{100}(160) = 72$

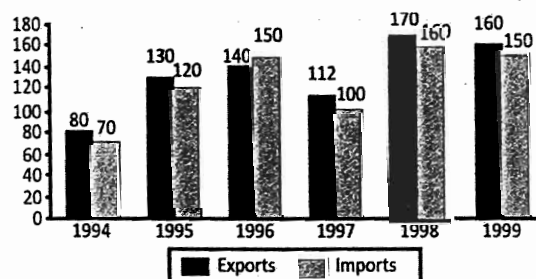
Required percentage

$$= \frac{70}{72} \times 100 = 97.2\%$$

Exercise -2

Bar Chart

These questions are based on the following graph:



1. In how many of the given years was the exports at least 10% more than the imports?

- (a.) 0
(b.) 1

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(c.) 2

(d.) 3

(e.) 4

Ans: c

Exp: In 1994, exports

$$= 80 > 70 + \frac{10}{100}(70) = 77$$

In 1995, exports

$$= 130 < 120 + \frac{10}{100}(120) = 132$$

In 1996, exports < imports

∴ We need not consider this year

In 1997, exports

$$= 112 > 100 + \frac{10}{100}(100) = 110$$

In 1998, exports

$$= 170 < 160 + \frac{10}{100}(160) = 176$$

In 1999, exports

$$= 160 < 150 + \frac{10}{100}(150) = 165$$

∴ The given condition was satisfied in two years.

2. What was the average exports for the given period (in '000 crores)?

(a.) 145

(b.) 132

(c.) 126

(d.) 119

(e.) 138

Ans: b

Exp: Average exports

$$= \frac{80+130+140+112+170+160}{6} = 132$$

3. From 1995 to 1999, in which year was the percentage growth in exports, when compared to the previous year, the highest?

- (a.) 1995
 (b.) 1996
 (c.) 1997
 (d.) 1998
 (e.) 1999

Ans: a

Exp: Export in a year exceeded that in the previous year in 1995, 1996 and 1998. Percentages by which exports in 1995, 1996 and 1998 exceed

the exports in the previous year were $\frac{50}{80}(100)\%$, $\frac{10}{130}(100)\%$ and

$\frac{58}{112}(100)$ respectively.

Only in 1995 was the growth more than 60%.

4. What is the simple average annual growth rate in the imports from 1994 to 1999?

- (a.) 15
 (b.) 18
 (c.) 19
 (d.) 21
 (e.) 23

Ans: e

Exp: Imports in 1994 (in '000 crores) = 70

Imports in 1999 (in '000 crores) = 150

Percentage growth

$$= \frac{150 - 70}{70} \times 100 = 115\%$$

$$\text{Average annual growth} = \frac{115}{5} = 23$$

5. Among the years in which the imports as well as exports exceed those in

the previous years, in how many years was the percentage increase in imports less than the percentage increase in exports?

- (a.) 0
- (b.) 1
- (c.) 2
- (d.) 3
- (e.) 4

Ans: a

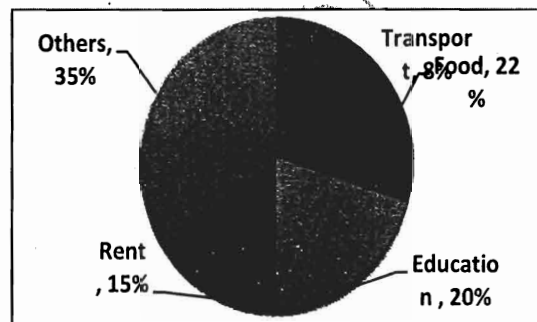
Exp: The imports as well as exports exceeded those in the previous years in 1995, 1996 and 1998.

∴ In none of the years was the given condition satisfied.

Exercise -3

Pie Charts

The following pie chart represents the break up of Raju's monthly expenses



1. If Raju spend Rs. 4500 more on food and transport together than he spent on rent, then find his monthly expenses.

- (a.) 15,000
- (b.) 25,000
- (c.) 30,000
- (d.) 35,000
- (e.) 40,000

Ans: c

Exp: Percentage of his expenditure spent on rent = 15%

Percentage of his expenditure spend on transport and food = 30%

∴ 30% - 15% = 15% = Rs. 4500

Monthly expenses = 100%

$$= \frac{100}{15}(4500) = \text{Rs. } 30,000$$

2. Raju spent 20% of his expenditure of 'others' on entertainment. This amounted to Rs. 2100. Find his expenditure on education.

- (a.) 4,500
(b.) 5,000
(c.) 6,500
(d.) 7,500
(e.) None of these

Ans: e

Exp: Expenditure on entertainment = 20% of 35% = 7% = 2100 1% = 300

Expenditure on education = 20% = Rs. 6,000.

3. Find the angle made by the expenditure on rent and 'others' put together.

- (a.) 120°
(b.) 160°
(c.) 180°
(d.) 210°
(e.) 240°

Ans: c

Exp: Total expenditure on rent and other
= 15% + 35% = 50%

$$\text{Required angle} = \frac{50}{100}(360^\circ) = 180^\circ$$

4. As prices dropped, Raju's expenditure on clothes dropped by 10%. As a result of this, his expenditure on 'others' decreased from Rs. 10,500 to Rs. 10,290. What percentage of his expenditure on 'others' was spent on clothes?

- (a.) 10%
(b.) 12%
(c.) 15%
(d.) 20%
(e.) 22.5%

Ans: d

Exp: Decrease in expenditure on 'others'
= 10500 - 10290
= Rs. 210

∴ Initial expenditure on clothes

Put Your Own Notes

$$= 210 \left(\frac{100}{10} \right) = \text{Rs. } 2100$$

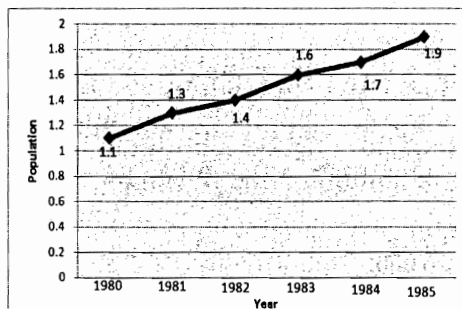
Required percentage

$$= \frac{2100}{10500} (100) = 20\%$$

Exercise-4

Line Graphs

The following line graph represents the population (in millions) of country X in each year from 1980 to 1985.



5. Find the percentage increase in the population of X from 1980 to 1985.

(a.) 60

(b.) $66\frac{2}{3}$

(c.) 70%

(d.) 75

(e.) None of these

Ans: e

Exp: Required percentage

$$= \frac{1.9 - 1.1}{1.1} (100)$$

$$= 72\frac{8}{11}\%$$

6. In which of the given years, from 1981 to 1985, did the population increase by the highest percentage over the previous years?

(a.) 1981

(b.) 1982

(c.) 1983

(d.) 1984

(e.) 1985

Ans: a

Exp: As the maximum increase in population between any two years is 0.2 million (1981, 1983 and 1985) and as the increase in population in 1981 is on a lower base, the percentage increase in 1981 would be the highest.

7. If the percentage of women in the population of country X in the years from 1980 to 1985 are 40, 45, 50, 55, 60 and 65 respectively, then in which of the given years when compared to its previous years, did the female population show the highest percentage increase?

(a.) 1985

(b.) 1984

(c.) 1983

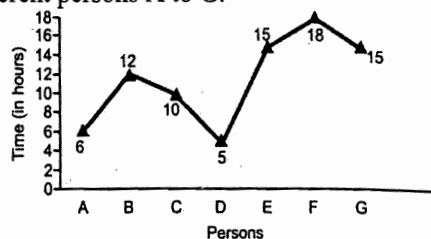
(d.) 1982

(e.) 1981

Ans: e

Exp: As the percentage increase in total population, and the percentage increase in the share of female population in 1981 (40-45, i.e., 12.5%) is the highest, the percentage increase in female population would be the highest in that year.

Direction for question 16 to 19: These questions are based on the following line graph, which shows the time taken to complete a particular task by seven different persons A to G.



8. In how much time can B and D together complete the task?

(a.) $5\frac{7}{17}$ hours

(b.) $5\frac{9}{17}$ hours

(c.) $3\frac{9}{17}$ hours

(d.) $3\frac{7}{17}$ hours

(e.) $4\frac{2}{17}$ hours

Ans: c

Put Your Own Notes

9. If C leaves 3 hours after A and C started the task, then how many more hours will A take to complete it?
- (a.) 1.2 hours
 - (b.) 2.4 hours
 - (c.) 1.5 hours
 - (d.) 2.5 hours
 - (e.) 3.8 hours

Ans: a

10. F, E and G worked on the task for one hour and then E left. If F and G together complete the remaining work, then how many more hours did F and G take to finish the remaining work?

- (a.) $6\frac{9}{11}$ hours
- (b.) $7\frac{7}{11}$ hours
- (c.) $5\frac{7}{11}$ hours
- (d.) $6\frac{7}{11}$ hours
- (e.) $4\frac{9}{11}$ hours

Ans: d

11. Approximately how long will it take for A, C, D and F together to complete the task?
- (a.) 1.5 hours
 - (b.) 1.7 hours
 - (c.) 2.3 hours
 - (d.) 2.1 hours
 - (e.) 1.9 hours

Ans: e

Put Your Own Notes

CHAPTER 9

AREA OF PLANE FIGURES

9.1 Triangles

The area of a triangle is represented by the symbol Δ . For any triangle, the three sides are represented by a , b and c and the angles opposite these sides represented by A , B and C respectively.

For any triangle in general,

- (i) When the measurements of three sides a , b , c are given,

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{where } s = \frac{a+b+c}{2}$$

This is called Hero's formula.

- (ii) When base (b) and altitude (height) to that base are given,

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{altitude} = \frac{1}{2} b.h$$

- (iii) $\text{Area} = \frac{1}{2} ab \cdot \sin C = \frac{1}{2} bc \cdot \sin A$

$$= \frac{1}{2} ca \cdot \sin B.$$

- (iv) $\text{Area} = \frac{abc}{4R}$ where R is the circum radius of the triangle.

- (v) $\text{Area} = r \cdot s$ where r is the in radius of the triangle and s , the semi-perimeter.

Out of these five formulae, the first and the second are the most commonly used and are also more important from the examination point of view.

- (vi) For a right angled triangle,

$$\text{Area} = \frac{1}{2} \times \text{Product of the sides containing the right angle.}$$

- (vii) For an equilateral triangle

$$\text{Area} = \frac{\sqrt{3} \cdot a^2}{4} \text{ where 'a' is the side of the triangle. The height of an}$$

$$\text{equilateral triangle } \frac{\sqrt{3} \cdot a}{4}.$$

- (viii) For an isosceles triangle

$$\text{Area} = \frac{b}{4} \sqrt{4a^2 - b^2} \text{ where 'a' is length of each of the two equal}$$

sides and b is the third side

9.2 Quadrilateral

- (i) For any quadrilateral

Area of the quadrilateral = $\frac{1}{2} \times \text{one diagonal} \times \text{sum of the offsets drawn to that diagonal.}$

- (ii) For a cyclic quadrilateral where the four sides measure a, b, c and d respectively.

$$\text{Area} = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

Where s is the semi-perimeter, i.e $s = \frac{(a+b+c+d)}{2}$

- (iii) For a trapezium

Area of a trapezium = $\frac{1}{2} \times \text{sum of parallel sides} \times \text{distance between them}$

$$= \frac{1}{2} \times (AD+BC) \times AE.$$

- (iv) For a parallelogram

Area = Base $\times$ Height

Area = Product of two sides sine of included angle.

- (v) For a rhombus

Area = $\frac{1}{2} \times \text{product of the diagonals}$

Perimeter = $4 \times \text{side of the rhombus}$

- (vi) For a rectangle

Area = Length $\times$ Breadth

Perimeter = $2(l+b)$, where l and b are the length and the breadth of the rectangle respectively.

- (vii) For a square

Area = Side²

Area = $\frac{1}{2} \times \text{Diagonal}^2$

[Where the diagonal = $\sqrt{2} \times \text{side}$]

Perimeter = $4 \times \text{Side.}$

Put Your Own Notes

(viii) For a polygon

Area of a regular polygon = $\frac{1}{2} \times \text{Perimeter} \times \text{Perpendicular distance from the centre of the polygon to any side.}$

For a polygon which is not regular, the area has to be found out by dividing the polygon into suitable number of quadrilaterals and triangles and adding up the areas of all such figure present in the polygon.

9.3 Circle

(i) Area of the circle = πr^2 where r is the radius of the circle.

Circumference = $2\pi r$

(ii) Sector of a circle

Length of arc = $\frac{\theta}{360^\circ} \times 2\pi r$

Area = $\frac{\theta}{360^\circ} \times \pi r^2$ where θ is the angle of the sector in degrees and r is the radius of the circle.

Area = $\left(\frac{1}{2}\right) l r$; l is length of arc and r is radius.

(iii) Ring: Ring is the space enclosed by two concentric circles.

Area = $\pi R^2 - \pi r^2 = \pi(R+r)(R-r)$ where R is the radius of the outer circle and r is the radius of the inner circle.

9.4 Ellipse

(i) Area = πab where " a " is semi-major and " b " is semi-minor axis.

(ii) Perimeter = $\pi(a+b)$

Put Your Own Notes

CHAPTER 10

AREA OF VOLUMES OF SOLIDS

- (i) Solids are three-dimensional objects which, in addition to area, have volume also. For solids, two different types of area are defined.
 - (a) Lateral surface area or curved surface area and
 - (b) Total surface area
- (ii) As the name itself indicates, lateral surface area is the area of the LATERAL surfaces of the solid. Total surface area includes the area of the top and the bottom surfaces also of the solid. Hence, total surface area = Lateral surface area + Area of the top face + Area of the bottom face.
- (iii) In solids (like cylinder, cone, sphere) where the lateral surface is curved, the lateral surface area is usually referred to as the "curved surface area".
- (iv) For any solid, whose faces are regular polygons, there is a definite relationship between the number of vertices, the number of sides and the number of edges of the solid. This relationship is given by "Euler's Rule".
- (v) Number of faces + Number of vertices = Number of edges + 2
 (Euler's Rule)

10.1 Prism

- (i) A right prism is a solid whose top and bottom faces (bottom face is called base) are parallel to each other and the identical polygons (of any number of sides) that are parallel. The faces joining the top and bottom faces are rectangles and are called lateral faces. There are as many lateral faces as there are sides in the base. The distance between the base and the top is called height or length of the right prism.
- (ii) In a right prism, if a perpendicular is drawn from the centre of the top face, it passes through the centre of the base.
- (iii) For any prism,

Lateral surface area = Perimeter of base $\times$ Height of the prism.

Total surface area = Lateral surface area + 2 $\times$ Area of base.

Volume = Area of base $\times$ Height of the prism.

10.2 Cuboid or Rectangular Solid

A right prism whose base is a rectangle is called a rectangular solid or cuboid. If l and b are respectively the length and breadth of the base and h , the height, then

$$\text{Volume} = lbh$$

$$\text{Lateral surface area} = 2(l + b) \cdot h$$

$$\text{Total surface area} = 2(l + b)h + 2lb$$

$$= 2(lb + lh + bh)$$

$$\text{Longest diagonal of the cuboid} = \sqrt{l^2 + b^2 + h^2}$$

10.3 Cube

- (i) A right prism whose base is a square and height is equal to the solid of the base is called a cube.

$$\text{Volume} = a^3 \text{ where } a \text{ is the edge of the cube}$$

$$\text{Lateral surface area} = 4a^2$$

$$\text{Total surface area} = 6a^2$$

- (ii) The longest diagonal of the cube (i.e., the line joining one vertex on the top face to the diagonally opposite vertex on the bottom face) is called the diagonal of the cube. The length of the diagonal of the cube is $a\sqrt{3}$.

10.4 Cylinder

- (i) A cylinder is equivalent to a right prism whose base is a circle. A cylinder has a single curved surface as its lateral faces. If r is the radius of the base and h is the height of the cylinder.

$$\text{Volume} = \pi r^2 h$$

$$\text{Curved surface area} = 2\pi rh$$

$$\text{Total surface area} = 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$$

- (ii) A hollow cylinder has a cross section of a ring. Volume of the material contained in a hollow cylindrical ring. $= \pi(R^2 - r^2)h$ where R is the outer radius, r is the inner radius and h , the height.

10.5 Pyramid

- (i) A solid whose base is a polygon and whose faces are triangles is called a pyramid. The triangular faces meet at a common point called

Put Your Own Notes

vertex. The perpendicular from the vertex to the base is called the height of the pyramid. A pyramid whose base is a regular polygon and the foot of the perpendicular from the vertex to the base coincides with the centre of the base, is called a right pyramid.

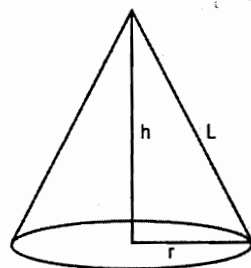
- (ii) The length of the perpendicular from the vertex to any side of the base (please note that this side will be the base of one of the triangular lateral faces of the prism) along the slant lateral surfaces is called the slant height of the prism.

$$\text{Volume of pyramid} = \frac{1}{2} \times \text{Area of base} \times \text{Height}.$$

$$\text{Lateral surface area} = \frac{1}{2} \times \text{Perimeter of the base} \times \text{Slant height}.$$

$$\text{Total surface area} = \text{Lateral surface area} + \text{Area of the base}.$$

10.6 Cone



- (i) A cone is equivalent to a right pyramid whose base is a circle. The lateral surface of a cone does not consist of triangles like in a right pyramid but is a single curved surface.
- (ii) If r is the radius of the base of the cone, h is height of the cone and l is the slant height of the cone, when we have the relationship.

$$l^2 = r^2 + h^2$$

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

$$\text{Curved surface area} = \pi r \cdot l$$

$$\text{Total surface area} = \pi r l + \pi r^2 = \pi r (l + r)$$

- (iii) A cone can be formed by taking the sector of a circle and joining together its straight edges. If the radius of the sector is R and the angle of the sector is θ° , then we have the following relationships between the length of the arc and area of the sector on the one hand and base perimeter of the cone and curved surface area of the cone on the other hand.

$$\text{Radius of the sector} = \text{Slant height of the cone i.e., } R = l$$

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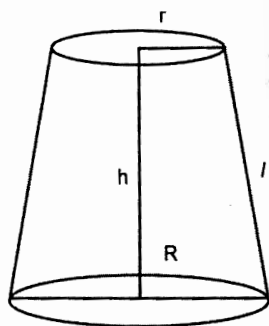
Length of the arc of the sector = Circumference of the base of the cone.

$$\text{i.e., } \frac{\theta}{360} \times 2 \times \pi R = 2\pi r$$

$$\Rightarrow r = \frac{\theta}{360} \times R$$

And area of the sector = curved surface area (actually, from this last equation, substituting the values from the first two equations, we can get the curved surface area of the cone, which is what is given previously as equal to $\pi r l$)

10.7 Cone Frustum



- (i) If a cone is cut into two parts by a plane parallel to the base, the portion that contains the base is called the frustum of a cone,
- (ii) If r is the top radius; R , the radius of the base; h the height and l the slant height of a frustum of a cone.

$$\text{Lateral surface area of the cone} = \pi(R + r)$$

$$\text{Total surface area} = \pi(R^2 + r^2 + R \cdot l + r \cdot l)$$

$$\text{Volume} = \frac{1}{3} \pi h (R^2 + Rr + r^2)$$

$$l^2 = (R - r)^2 + h^2$$

If H is the height of the complete cone from which the frustum is cut, then from similar triangles, we can write the following relationship.

$$\frac{r}{R} = \frac{H - h}{H}$$

A bucket that is normally used in a house is a good example of the frustum of a cone. The bucket is actually the inverted form of the frustum.

10.8 Frustum of a Pyramid

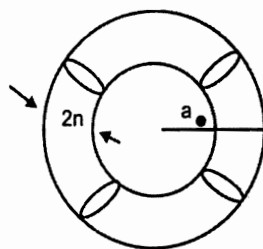
- (i) A pyramid left after cutting of a portion at the top by a plane parallel to the base is called a frustum of a pyramid.
- (ii) If A_1 is the area of the base; A_2 the area of the top and h , the height of the frustum,

$$\text{Volume of frustum} = \frac{1}{3} \times h \times (A_1 + A_2 + \sqrt{A_1 A_2})$$

$$\text{Lateral surface area} = \frac{1}{2} \times (\text{Sum of perimeters of base and top}) \times \text{Slant height}$$

$$\text{Total surface area} = \text{Lateral surface area} + A_1 + A_2$$

10.9 Torus



- (i) A torus is a three-dimensional figure produced by the revolution of a circle about an axis laying in its plane but not intersecting the circle. The shape of the rubber tube in a bicycle (when it is inflated fully) is an example of a torus. If r is a radius of the circle that rotates and a is the distance between the centre of the circle and the axis of revolution.
- (ii) Surface area of the torus $= 4\pi^2 ra$
- (iii) Volume of the torus $= 2\pi^2 r^2 a$

10.10 Sphere

- (i) Any point on the surface of a sphere is equidistant from the centre of the sphere. This distance is the radius of the sphere.
- (ii) Surface area of a sphere $= 4\pi r^2$
- (iii) Volume of a sphere $= \left(\frac{4}{3}\right)\pi r^3$
- (iv) The curved surface area of a hemisphere is equal to half the surface area of a sphere, i.e., $2\pi r^2$.

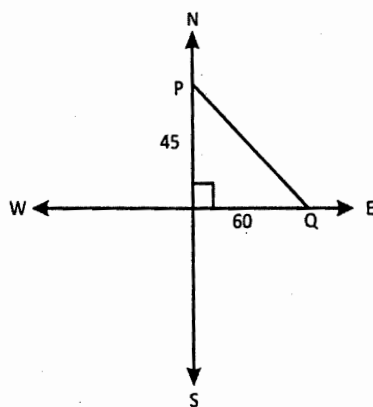
Examples:

- (a) P and Q leave a point at the same time. P travelled towards North at a speed of 15 km/hr. Q travelled towards west at a speed of 20 km/hr.

Put Your Own Notes

Find the shortest distance between them after 3 hours.

Exp: The position of P and Q after 3 hours.



In 3 hours, P would have travelled 45 km and Q would have travelled 60 km. Shortest distance between them $(PQ) = \sqrt{45^2 + 60^2} = 75 \text{ km}$.

- (b) The wheel of a motorcar makes 800 revolutions in covering 880 m. Find its diameter (Take $\pi = \frac{22}{7}$).

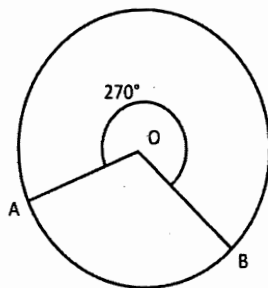
Exp: Let the diameter be dm .

Distance covered = (Number of revolutions) (circumference of the wheel)

$$\therefore 880 = (800) \left(\frac{22}{7} \right) (d)$$

$$d = 0.35$$

(c)



In the figure above, O is the centre of the circle. Find the length of the mirror arc AB if $OA = 7 \text{ cm}$. (Take $\pi = \frac{22}{7}$)

Exp: $\angle AOB = 360^\circ - 270^\circ = 90^\circ$

$$\text{Length of the mirror arc } AB = \frac{\theta}{360^\circ} \times 2\pi r$$

Put Your Own Notes

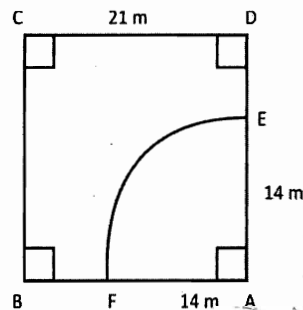
$$= \left(\frac{90^\circ}{360^\circ} \right) \left((2) \left(\frac{22}{7} \right) (7) \right)$$

$$= 11 \text{ cm}$$

- (d) A goat is tied to a corner of a square field of side 21 m with a rope of length 14 m. Find the area of the square field that the goat cannot graze.

(Take $\pi = \frac{22}{7}$).

Exp:



Area that the goat cannot graze = (Area of ABCD) - (Area of sector AEF)

$$= (21^2) - \left(\frac{90^\circ}{360^\circ} \pi (14)^2 \right)$$

$$= 287 \text{ sq.m}$$

- (e) A circular garden has a radius of 15 m. It is surrounded by a circular path of width 7 m. If the path is to be polished at a rate of Rs. 15 per sq. m., find the total cost of polishing the path. (Take $\pi = \frac{22}{7}$)

Exp: Total cost of polishing = (15) (Area to be covered)

$$= 15 \left(\frac{22}{7} (22^2 - 15^2) \right)$$

$$= \text{Rs. } 12210$$

- (f) Find the volume of the largest right circular cylinder which can be cut from a cube of side 7 cm. (Take $\pi = \frac{22}{7}$)

Exp: The largest right circular cylinder will have its height as well as its diameter equal to the side of the cube.

$\therefore$ Its volume

$$= \frac{22}{7} \left(\left(\frac{7}{2} \right)^2 \right) (7) = 269.5 \text{ cubic cm}$$

- (g) A roller has a length of 3 m. Its diameter is 0.7 m. It requires 600 revolutions of the roller to level a road. The cost of usage of the roller is Rs. 5 per sq. m. Find the total cost of leveling the road. (Take $\pi = \frac{22}{7}$)

Put Your Own Notes

Exp: Curved surface area of the roller

$$= 2 \left(\frac{22}{7} \right) \left(\frac{0.7}{2} \right) (3) = 6.6 \text{ sq. m}$$

Cost of leveling the road

$$= (600) (6.5) (5)$$

$$= \text{Rs. } 19800.$$

- (h) A cylindrical vessel of base radius 4 cm is filled with water to a height of 6 cm. If lead shots each of radius 2 mm are dropped into it and the water level rises to 8.50 cm. Find the number of lead shots dropped.

- (i) 500
- (ii) 3750
- (iii) 1000
- (iv) 1500

Ans: b

- (i) Three cubes whose diagonals are $6\sqrt{3} \text{ cm}$, $8\sqrt{3} \text{ cm}$ and $10\sqrt{3} \text{ cm}$ are melted to form a new cube. Find the length of the diagonal of the cube formed.

- (i) $11\sqrt{3} \text{ cm}$
- (ii) $12\sqrt{3} \text{ cm}$
- (iii) $14\sqrt{3} \text{ cm}$
- (iv) $13\sqrt{3} \text{ cm}$

Ans: b

- (j) The thickness of a pipe is 2 mm and its external diameter is 0.8 cm. If 1 cm^3 of the material of the pipe weighs 15 gm, then find the weight of the pipe of length 28 cm.

- (i) 25.6 gm
- (ii) 31.8 gm
- (iii) 158.4 gm
- (iv) 63.6 gm

Ans: b

- (k) A goat each is tethered with ropes of length 14 m, to two diagonally opposite corners of a rectangular plot with dimensions $35 \text{ m} \times 14 \text{ m}$. Find the total area that they can graze in the rectangular plot.

- (i) 182 sq.m
- (ii) 192 sq.m
- (iii) 308 sq.m
- (iv) 154 sq.m

Ans. c

- (l) The area of a circle, a square and an equilateral triangle are equal. If the perimeters of the circle, the square and the triangle are C, S and T respectively. Which of the following holds true?

Put Your Own Notes

- (i) $C < T < S$
- (ii) $S < T < C$
- (iii) $C < S < T$
- (iv) $T < C < S$

Ans: c

- (m) If a spherical balloon is inflated in such a way that its radius becomes thrice, by how many times will its surface area increase, when compared to its original surface area?

- (i) 3 times
- (ii) 8 times
- (iii) 27 times
- (iv) 81 times

Ans: b

- (n) The inside of a well thirty feet deep was given a finishing of brick-work. If the inner diameter of the well after the brick-work, which was six inches thick, was completed was ten feet, what was the volume of the brick-work?

- (i) 167.5π cubic feet
- (ii) 157.5π cubic feet
- (iii) 187.5π cubic feet
- (iv) 207.5π cubic feet

Ans. b

- (o) The cost of paving a floor with square tiles at Rs. 7 per sq. m. is Rs. 2240. If the length of the floor is twice the breadth, find the perimeter of the floor.

- (i) $64\sqrt{10} \text{ m}$
- (ii) $84\sqrt{10} \text{ m}$
- (iii) $8\sqrt{10} \text{ m}$
- (iv) $84\sqrt{10} \text{ m}$

Ans. d

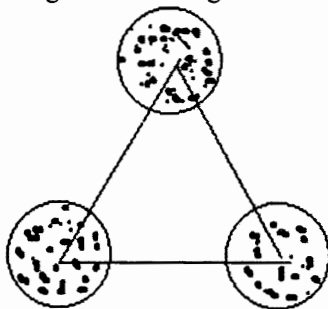
ASSIGNMENT SHEET -1

- Three boxes are coloured red, blue and green and so are three balls. In how many ways can one put the balls one in each box such that no ball goes into the box of its own colour?
 - 1
 - 2
 - 3
 - 4
- Write $d=1$ degree, $r=1$ radian and $g=1$ grad. Then which of the following is true? (100 grad = a right angle)
 - $\cos d < \cos r < \cos g$
 - $\cos r < \cos g < \cos d$
 - $\cos r < \cos d < \cos g$
 - $\cos g < \cos d < \cos r$
- The base diameter of a glass is 20% smaller than the diameter at the rim. The glass is filled to half the height. The ratio of empty to filled volume of the glass is
 - $\frac{\sqrt{10}-\sqrt{9}}{\sqrt{9}-\sqrt{8}}$
 - $\frac{10-9}{9-8}$
 - $\frac{10^2-9^2}{9-8}$
 - $\frac{10^3-9^3}{9^3-8^3}$
- A circle drawn in the x - y coordinate plane passes through the origin and has chords of lengths 8 units and 7 units on the x and y axes, respectively. The coordinates of its centre are
 - (8,7)
 - (-8,7)
 - (-4,3.5)
 - (4,3.5)
- The number of diagonals of a convex deodecagon (12-gon) is
 - 66
 - 54
 - 55
 - 60
- A wheel barrow with unit spacing between its wheels is pushed along a semi-circular path of mean radius 10. The difference between distances covered by the inner and outer wheels is
 - 0
 - 10
 - π
 - 2π
- A vendor sells articles having a cost price of Rs. 100 each. He sells these articles at a premium price during first eight months, and at a sale price, which is half of the premium price, during next four months. He makes a net profit of 20% at the end of the year. Assuming that equal numbers of articles are sold each month, what is the premium price of the article?
 - 122
 - 144
 - 150
 - 160
- The missing number is

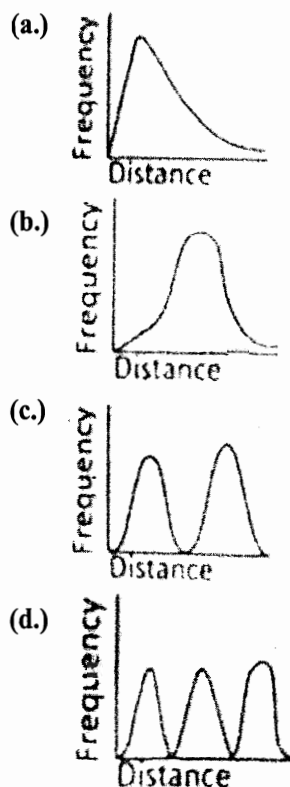
5			
8		2	
7	9		-5
5	9	9	?

 - 19
 - 5
 - 9
 - 9

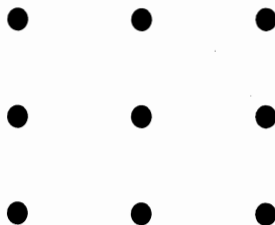
9. Three circles of equal diameters are placed such that their centres make an equilateral triangles as in the figure



Within each circle, 50 points are randomly scattered. The frequency distribution of distances between all possible pairs of points will look as



10.



The minimum number of straight lines required to connect the nine points above without lifting the pen or retracing is

- (a.) 3
 (b.) 4
 (c.) 5
 (d.) 6

11. Suppose three meetings of a group of professors were arranged in Mumbai, Delhi and Chennai. Each professor of the group attended exactly two meetings. 21 professors attended Mumbai meeting, 27 attended Delhi meeting and 30 attended Chennai meeting. How many of them attended both the Chennai and Delhi meetings?

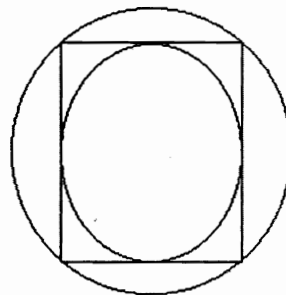
- (a.) 18
 (b.) 24
 (c.) 26
 (d.) Cannot be found from the above information

12. Decode

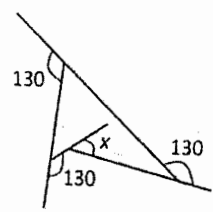
G E N T S T U
 I S S O L V D
 L I I S P A E
 L M H T R B N
 E E L B O L T
 T N I Y B E S

- (a.) GENT STUDENTS CAUSE LITTLE HEARTS BURNS
 (b.) STUDENTS ARE INTELLIGENT BUT PROBLEM IS NOT SOLVABLE
 (c.) THIS PROBLEM IS UNSOLVABLE BY ANY STUDENT
 (d.) THIS PROBLEM IS SOLVABLE BY INTELLIGENT STUDENTS

13. There is an inner circle and an outer circle around a square. What is the ratio of the area of the outer circle to that of the inner circle?



- (a.) $\sqrt{2}$
 (b.) 2
 (c.) $2\sqrt{2}$
 (d.) $\sqrt{3/2}$

14. One is required to tile a plane with congruent regular polygons. With which of the following polygons is this possible?
- 6-gon
 - 8-gon
 - 10-gon
 - 12-gon
15. Most Indian tropical fruit trees produce fruits in April-May. The best possible explanation for this is
- optimum water availability for fruit production.
 - the heat allows quicker ripening of fruit.
 - animals have no other source of food in summer.
 - the impending monsoon provides optimum conditions for propagation.
16. The probability that a ticketless traveler is caught during a trip is 0.1. If the traveler makes 4 trips, the probability that he/she will be caught during at least one of the trips is:
- $1-(0.9)^4$
 - $(1-0.9)^4$
 - $1-(1-0.9)^4$
 - $(0.9)^4$
17. Let A, B be the ends of the longest diagonal of the unit cube. The length of the shortest path from A to B along the surface is
- $\sqrt{3}$
 - $1+\sqrt{2}$
 - $\sqrt{5}$
 - 3
18. The statement: "The father of my son is the only child of your parents"
- can never be true
 - is true in only one type of relation
 - can be true for more than one type of relations
 - can be true only in a polygamous family
19. How many digits are there in 3^{16} when it is expressed in the decimal form?
- Three
 - Six
 - Seven
 - Eight
20. "The clue is hidden in this statement", read the note handed to Sherlock by Moriarty, who hid the stolen treasure in one of the ten pillars. Which pillar is it?
- X
 - II
 - III
 - IX
21. What is angle x in the schematic diagram given below?
- 
- The diagram shows a triangle with three interior angles, each labeled 130. An exterior angle at one of the vertices is labeled x . The exterior angle is formed by extending one of the sides of the triangle.
- 60
 - 50
 - 40
 - 30
22. A 3m long car goes past a 4m long truck at rest on the road. The speed of the car is 7 m/s. The time taken to go past is
- $4/7$ s
 - 1 s
 - $7/4$ s
 - $10/7$ s
23. Of three persons A, B and C , one always lies while the others always speak the truth. C asked A , "Do you always speak the truth, yes or no?" He said something that C could not hear. So, C asked B , "what did A say?" B replied, " A said No". So, who is the liar?
- A
 - B
 - C
 - Cannot be determined

24. Consider 3 parallel strips of 10 m width running around the Earth, parallel to the equator; A_1 at the Equator, A_2 at the Tropic of Cancer and A_3 at the Arctic Circle. The order of the areas of the strips is
- $A_1 < A_2 < A_3$
 - $A_1 = A_2 > A_3$
 - $A_1 > A_2 = A_3$
 - $A_1 > A_2 > A_3$
25. A pyramid shaped toy is made by tightly placing cubic blocks of $1 \times 1 \times 1 \text{ cm}^3$. The base of toy is a square $4 \times 4 \text{ cm}^2$. The width of each step is 0.5 cm . How many blocks are required to make the toy?
- 30
 - 34
 - 36
 - 40
26. A single celled spherical organism contains 70% water by volume. If it losses 10% of its water content, how much would its surface area change by approximately?
- 3%
 - 5%
 - 6%
 - 7%
27. Two plane mirrors facing each other are kept at 60° to each other. A point is located on the angle bisector. The number of images of the point is
- 6
 - 3
 - 5
 - Infinite
28. Suppose
- $x = 4$
 - Then $x - 4 = x^2 - 4^2$ (as both sides are zero)
 - Therefore $(x - 4) = (x - 4)(x + 4)$
Cancelling $(x - 4)$ from both sides
 - $1 = (x + 4)$
 - Then $x = -3$
Which is the wrong step?
- 1 to 2
 - 2 to 3
 - 3 to 4
 - 4 to 5
29. Let m and n be two positive integers such that $m + n + mn = 118$. Then the value of $m + n$ is
- Not uniquely determined
 - 18
 - 20
 - 22
30. From a group of 40 players, a cricket team of 11 players is chosen. then, one of the eleven is chosen as the captain of the team. The total number of ways this can be done is $\left[\begin{matrix} m \\ n \end{matrix} \right]$ below means the number of ways n objects can be chosen from m objects]
- $\begin{pmatrix} 40 \\ 11 \end{pmatrix}$
 - $11 \begin{pmatrix} 40 \\ 11 \end{pmatrix}$
 - $29 \begin{pmatrix} 40 \\ 11 \end{pmatrix}$
 - $\begin{pmatrix} 39 \\ 10 \end{pmatrix}$
31. I bought a shirt at 10% discount and sold it to a friend at a loss of 10%. If the friend paid me Rs. 729.00 for the shirt, what was the undiscounted price of the shirt?
- Rs. 900
 - Rs. 800
 - Rs. 1000
 - Rs. 911.25
32. Jar W contains 40 whit marbles and jar B contains 40 black marbles. Ten black marbles from B are transferred to W and mixed thoroughly. Now, ten randomly selected marbles from W are put back in Jar B to make 40 marbles in each jar. The number of black marbles in W
- would be equal to the number of white marbles in B
 - would be more than the number of white marbles in B
 - would be less than the number of white marbles in B
 - cannot be determined from the information given

33. How many non-negative integers less than 10,000 are there such that the sum of the digits of the number is divisible by three?

(a.) 1112
(b.) 2213
(c.) 2223
(d.) 3334

34. If $aN \Rightarrow S$
 $eF \Rightarrow I$
 $gH \Rightarrow M$

then $nS \Rightarrow ?$

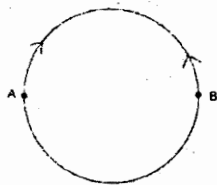
(a.) T
(b.) A
(c.) L
(d.) K

35. In each of the following groups of words is a hidden number, based on which you should arrange them in ascending order. Pick the right answer:

A. Tinsel event
B. Man in England
C. Good height
D. Last encounter

(a.) A, B, C, D
(b.) C, B, D, A
(c.) A, C, D, B
(d.) C, D, B, A

36. Two ants, initially at diametrically opposite points A and B on a circular ring of radius R , start crawling towards each other. The speed of the one at A is half of that of the one at B . The point at which they meet is at a straight line distance of



(a.) R from A
(b.) $\frac{3R}{2}$ from A
(c.) R from B
(d.) $\frac{3R}{2}$ from B

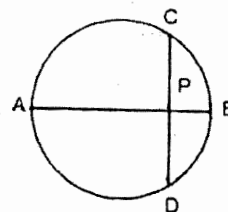
37. Starting from a point A you fly one mile south, then one mile east, then one mile north which brings you back to point A . Point A is NOT the north pole. Which of the following MUST be true?

(a.) You are in the Northern Hemisphere
(b.) You are in the Eastern Hemisphere
(c.) You are in the Western Hemisphere
(d.) You are in the Southern Hemisphere

38. Information in DNA is in the form of sequence of 4 bases namely A, T, G and C . The proportion of G is the same as that of C , and that of A is the same as that of T . Which of the following strands of DNA will potentially have maximum diversity (i.e., maximum information content per base)?

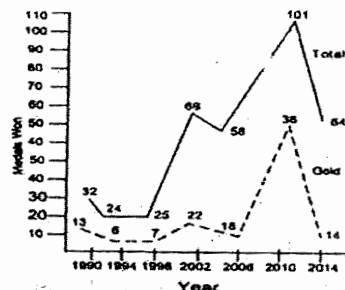
(a.) length 1000 bases with 10% G
(b.) length 2000 bases with 10% A
(c.) length 2000 bases with 40% T
(d.) length 1000 bases with 25% C

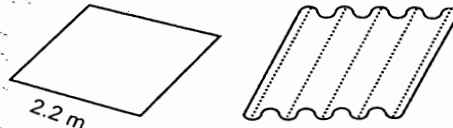
39. AB is the diameter of a circle. The chord CD is perpendicular to AB intersecting it at P . If $CP = 2$ and $PB = 1$, the radius of the circle is



(a.) 1
(b.) 2.5
(c.) 2
(d.) 5

40. Based on the graph, which of the following statements is NOT true?

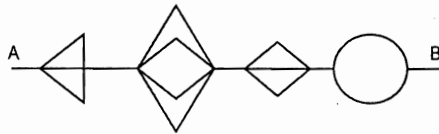


- (a) Number of gold medals increased whenever total number of medals increased
- (b) Percentage increase in gold medals in 2010 over 2006 is more than the corresponding increase in total medals
- (c) Every time non-gold medals together account for more than 50% of the total medals
- (d) Percentage increase in gold medals in 2010 over 2006 is more than the corresponding increase in 2002 over 1998
41. We define a function $f(N)$ = of digit of N , expressed as decimal number. $g.f(137) = 1+3+7 = 11$. Evaluate $f(2^7 3^5 5^6)$
- (a.) 10
(b.) 18
(c.) 28
(d.) 11
42. Every month the price of a particular commodity falls in this order: 1024, 640, 400, 250, ...
What is the next value?
- (a.) 156.25
(b.) Approximately 39
(c.) 64
(d.) 40
43. A cubical piece of wood was filed to make it into the largest possible sphere. What fraction of the original volume was removed?
- (a.) More than $3/4$
(b.) $1/2$
(c.) Slightly less than $1/2$
(d.) Slightly more than $1/2$
44. Three volumes of a Hindi book, identical in shape and size, are next to each other in a shelf, all upright so that their spines are visible m let to right: I, II, III. A worm starts eating from the outside front cover of volume I, and eats its way horizontally to the outside back cover of volume I, and eats its way horizontally to the outside back cover of volume III. What is the distance travelled by the worm, if each volumes is 6 cm thick?
- (a.) 6 cm
(b.) 12 cm
(c.) 18 cm
(d.) A little more than 18 cm
45. What is the area of the triangle bounded by the lines $y = 2x$, $y = -2x$ and $y = 6$?
- (a.) 36
(b.) 18
(c.) 12
(d.) 24
46. A 2.2 m wide rectangular steel plate is corrugation as shown in the diagram. Each corrugation is a semi-circle in cross section having a diameter of cm. what will be the width of steel sheet after it is corrugated?
- 
- (a.) 1.4 m
(b.) 1.6.m
(c.) 0.7 m
(d.) 1.1 m
47. Ajay ,Bunty, Chinu and Deb were agent, backer, compounder and designer, but not necessarily in that order. Deb told the backer that Chinu is on his way. Ajay is sitting across the designer and next to the compounder. The designer didn't say anything. What is each person's occupation?
- (a.) Ajay -compounder, bunty-designer, chinu-backer ; Deb-agent
(b.) Ajay-compounder; Bunty -backer Chinu-agent; Deb -designer
(c.) Ajay-backer; bunty-agen; Chinu-designer; Deb-compounder
(d.) Ajay-backer ; Bunty-designer; Chinu-agent; Deb-compounder
48. Two platforms are separated horizontally by distance A and vertically by distance B . they are to be connected by a staircase having identical steps. If the minimum permissible step length is a and the maximum permissible step height is b , the number of steps the staircase can have is
- (a.) $\geq B/b$
(b.) $\leq A/a$
(c.) $\geq B/b$ and $\leq A/a$
(d.) $\leq B/b$ and $\geq A/a$

49. Then sum of first n natural numbers with one of them missed is 42. What is the number that was missed?

(a.) 1
 (b.) 2
 (c.) 3
 (d.) 4

50. A mouse has to go from point A to B without retracing any part of the path, and never moving backwards. What is the total numbers of distance paths that the mouse may take to go from A to B?



(a.) 11
 (b.) 48
 (c.) 72
 (d.) 24

51. A certain day, which is x days before 17th August is such that 50 days prior to that day, it was $4x$ days since March 30th of the same year. What is x ?

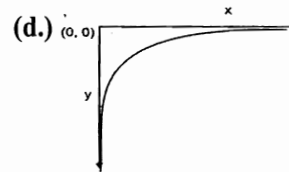
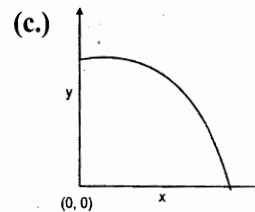
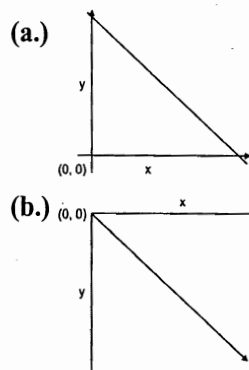
(a.) 18
 (b.) 30
 (c.) 22
 (d.) 16

52. What is the next term in the following sequence?

7, 11, 13, 17, 19, 23, 29,

(a.) 37
 (b.) 35
 (c.) 31
 (d.) 33

53. Which of the following figures best shows that y is inversely proportional to x ?



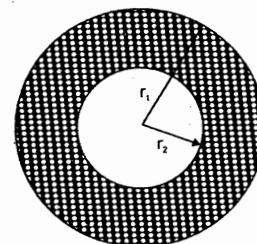
54. What is the maximum number of whole laddoos having diameter of 6 cm that can be packed in a box whose inner dimensions are $24 \times 18 \times 17 \text{ cm}^3$?

(a.) 24
 (b.) 30
 (c.) 33
 (d.) 36

55. If N, E and T are distinct positive integers such that $N \times E \times T = 2013$ then which of the following is the maximum possible sum of N, E and T ?

(a.) 3
 (b.) 2015
 (c.) 675
 (d.) 671

56. The areas of the inner circle and the shaded ring are equal. The radii r_1 and r_2 are related by



(a.) $r_1 = r_2$
 (b.) $r_1 = r_2 \sqrt{2}$
 (c.) $r_1 = r_2 \sqrt{3}$
 (d.) $r_1 = 2r_2$

57. The equation $m^2 - 33n + 1 = 0$, where m & n are integers, has

(a.) No solution
(b.) Exactly one solution
(c.) Exactly two solution
(d.) Infinitely many solution

58. Which of the following numbers is a perfect square?

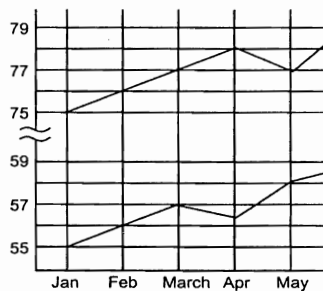
(a.) 1022121
(b.) 2042122
(c.) 3063126
(d.) 4083128

59. What is the 94th term of the following sequence

1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4,

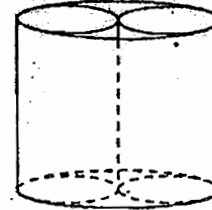
(a.) 8
(b.) 9
(c.) 10
(d.) 11

60. The following graphs depict variation in the value of Dollar and Euro in terms of the Rupee over six months.



Which of the following statements is true?

- (a.) Values of Dollar and Euro rose steadily from to june
(b.) Values of Dollar and Euro rose by equal rate between January to march
(c.) The rise in the value of Dollar from April to May is three times the fall in Euro during the same period
(d.) Values of Dollar and Euro rose equally between May and june
61. Two identical cylinders of the same height as a bigger hollow cylinder were put vertically into the latter to fit exactly into it. The volume of the bigger cylinder is V . The volume of each of the smaller cylinders is



(a.) $V/8$
(b.) $V/4$
(c.) $V/2$
(d.) V

62. What is the length of the longest rod that can be put in a hemispherical bowl of radius 10 cm such that no end of the rod is outside the bowl? (Assume that the rod has negligible thickness).

(a.) $10\sqrt{2}$ cm
(b.) $10\sqrt{3}$ cm
(c.) $10\sqrt{4}$ cm
(d.) $10\sqrt{5}$ cm

63. On a semi-circle of diameter 10m drawn on a horizontal ground are standing 4 boys A, B, C and D with distances $AB = BC = CD$. The length of line-segment joining A and B is



(a.) 5 m
(b.) 6 m
(c.) 7 m
(d.) $\frac{5\pi}{3}$ m

64. You get 20% returns on your investment annually, but also pay a 20% tax on the gain. At the end of 5 years, the net gain made by you (as percentage of the capital) is approximately

(a.) 0
(b.) 16
(c.) 80
(d.) 100

65. A cubic cavity of edge $20 \mu\text{m}$ is filled with a fluid with a cubic solid of edge $2 \mu\text{m}$. What percent of the cavity volume is occupied by the fluid?

(a.) 10.0
(b.) 20.0
(c.) 90.0
(d.) 99.9

66. The following table shows the price of diamond crystals of particular quality.

Wt. of a diamond crystal (in carat)	Price per carat (in lakh Rs.)
1	4
2	8
3	12
4	16

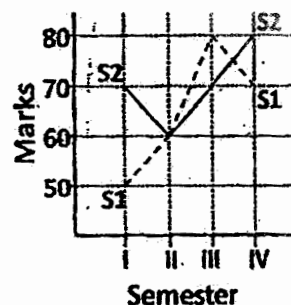
What will be the price (in lakh Rs.) of a 2.5 carat diamond crystal?

(a.) 10
(b.) 20
(c.) 25
(d.) 50

67. A man on the equator moves along 0° longitude up to 45°N . He then turns east and moves up to 90°E , and returns to the equator along 90°E . The distance covered in multiples of Earth's radius R is

(a.) $\left(\frac{3}{4}\pi\right)R$
(b.) $\left(\frac{\pi}{2} + \frac{\pi}{4\sqrt{2}}\right)R$
(c.) $\left(\frac{\pi}{2} + \frac{\pi}{2\sqrt{2}}\right)R$
(d.) $\left(\frac{\pi}{4} + \frac{\pi}{\sqrt{2}}\right)R$

68. Marks obtained by two students S1 and S2 in a four semester course are plotted in the following graph



Which of the following statements is true?

- (a.) S2 got higher marks than S1 in all four semesters
(b.) Over four semesters, S1 improved by a higher percentage compared to S2.
(c.) Total marks of S1 and S2 are equal
(d.) S1 and S2 did not get the same marks in any semester.

69. To go from the engine to the last coach of his train of length 200 m, a man jumped from his train to another train moving on a parallel track in the opposite direction, waited till the last coach of his original train appeared and then jumped back. In how much time did he reach the last coach if the speed of each train was 60 km/hr?

(a.) 5 s
(b.) 6 s
(c.) 10 s
(d.) 12 s

70. How many digits are there in $2^{17} \times 3^2 \times 5^{14} \times 7$?

(a.) 14
(b.) 15
(c.) 16
(d.) 17

71. The following sum is

$$1 + 1 - 2 + 3 - 4 + 5 - 6 \dots - 20 = ?$$

(a.) 10
(b.) -10
(c.) -11
(d.) -9

72. What is the next number in the following sequence?

2, 3, 5, 6, 3, 4, 7, 12, 4, 5, 9, ...

- (a.) 10
(b.) 20
(c.) 13
(d.) 6

73. After giving 20% discount on the marked price to a customer, the seller's profit was 20%. Which of the following is true?

(a.) $\text{Sale price} = \frac{\text{Marked price} + \text{Cost price}}{2}$

(b.) $\text{Sale price} < \frac{\text{Marked price} + \text{Cost price}}{2}$

(c.) $\frac{2}{3}(\text{Marked price} + \text{Cost price}) >$

$\text{Sale price} > \frac{\text{Marked price} + \text{Cost price}}{2}$

(d.) $\text{Sale price} > \frac{2}{3}(\text{Marked price} + \text{Cost price})$

74. Coordinates of a point in x, y, z space is (1, 2, 3). What would be the coordinates of its reflection in a mirror along the x, z plane?

- (a.) (-1, -2, -3)
(b.) (1, -2, -3)
(c.) (1, -2, 3)
(d.) (-1, 2, 3)

75. You are given 100 verbs using which you have to form sentences containing at least one verb, without repeating the verbs, under the condition that the number of verbs (from this set of 100) in any two sentences should not be equal. The maximum number of sentences you can form is

- (a.) 10
(b.) 13
(c.) 14
(d.) 100

76. A 4×4 magic square is given below:

1	15	14	4
12	6	7	9
8	10	11	5
13	3	2	16

How many 2×2 squares are there in it whose elements add up to 34?

- (a.) 6
(b.) 9
(c.) 4
(d.) 5

77. November 9, 1994 was a Wednesday. Then which of the following is true?

- (a.) November 9, 1965 is a Wednesday and November 9, 1970 is a Wednesday.
(b.) November 9, 1965 is not a Wednesday and November 9, 1970 is a Wednesday.
(c.) November 9, 1965 is a Wednesday and November 9, 1970 is not a Wednesday.
(d.) November 9, 1965 is not a Wednesday and November 9, 1970 is not a Wednesday.

78. If a 4 digit year (e.g. 1927) is chosen randomly, what is the probability that it is NOT a leap year?

- (a.) $\frac{3}{4}$
(b.) $\frac{1}{4}$
(c.) $< \frac{1}{4}$
(d.) $> \frac{3}{4}$

79. Three years ago, the difference in the ages of two brothers was 2 years. The sum of their present ages will double in 10 years. What is the present age of the elder brother?

- (a.) 6
(b.) 11
(c.) 7
(d.) 9

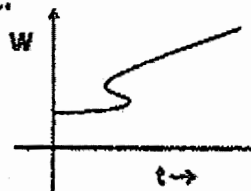
80. Find the missing number in the sequence

61, 52, 63, 94, ..., 18, 001, 121

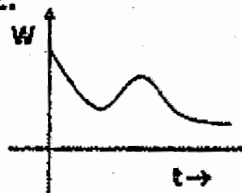
- (a.) 46
(b.) 70
(c.) 66
(d.) 4

81. If we plot the weight (w) versus get (t) of a child in a graph, the one that will never be obtained from amongst the four graphs given below is

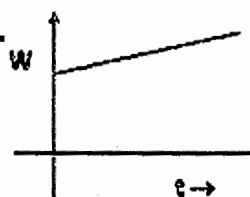
(a) 1.



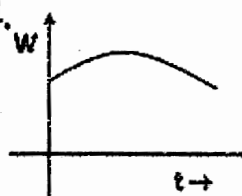
(b) 2.



(c) 3.



(d) 4.



82. A hemispherical bowl is being filled with water at constant volumetric rate. The level of water in the bowl increases

- (a) in direct proportion to time,
- (b) in inverse proportion to time,
- (c) faster than direct proportion to time,
- (d) slower than direct proportion to time.

83. Equal masses of two liquids of densities kg/m^3 and 4 kg/m^3 are missed thoroughly. The density of the mixture is

- (a) 4.8 kg/m^3
- (b) 5.0 kg/m^3
- (c) 5.2 kg/m^3
- (d) 5.4 kg/m^3

84. Two points A and B on the surface of the Earth have the following latitude and longitude co-ordinates.

A: 30° N , 45° E

B: 30° N , 135° W

If R is the radius of the Earth, the length of the shortest path from A to B is

(a) $\frac{\sqrt{3}}{2} \pi R$

(b) $\frac{\pi R}{3}$

(c) $\frac{\pi R}{6}$

(d) $\frac{2\pi R}{3}$

85. Amoebae are known to double in 3 min. Two identical vessels A & B, respectively contain one and two amoebae to start with. The vessel B gets filled in 3 hours. When will A get filled?

- (a) 3 hours
- (b) 2 hours 57 min
- (c) 3 hours 3 min
- (d) 6 hours

86. Students of a school are divided into 4 groups. What is the probability that three friends get into the same group?

(a) $\frac{3}{4}$

(b) $\frac{1}{64}$

(c) $\frac{1}{16}$

(d) $\frac{1}{3}$

87. A fruit vendor buys 120 Shimla apples at 4 for Rs. 100, and 120 Golden apples at 6 for Rs. 100. She decides to mix them and sell at 10 for Rs. 200. She will make

- (a) no profit, no loss
- (b) a loss of 4%
- (c) a gain of 4%
- (d) a loss of 10%

88. $4^0 + 4^2 + 4^{-2} + 4^{1/2} + 4^{-1/2} =$

(a) 4^0

(b) $4^{\frac{1}{2}} + 4^{-\frac{1}{2}}$

(c) $19\frac{9}{16}$

(d) $22\frac{9}{16}$

89. In an enclosure there were both crows and cows. If there were 30 heads and 100 legs, what fraction of them are crows?

(a.) $\frac{1}{3}$
(b.) $\frac{1}{4}$
(c.) $\frac{1}{10}$
(d.) $\frac{3}{10}$

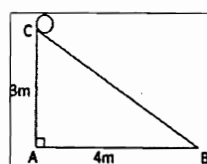
90. A cart wheel rolls along a straight line. If the distance covered is equal to the diameter of the wheel, what is the angle through which the wheel has turned?

(a.) 90°
(b.) between 90° and 120°
(c.) between 120° and 150°
(d.) between 150° and 180°

91. In a class of 10 students, 3 failed in 13. History, 6 failed in Geography and 2 failed in both. How many passed in both the subject?

(a.) 1
(b.) 2
(c.) 3
(d.) 0

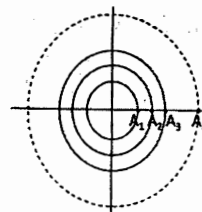
92.



As shown in the diagram above, a sphere is placed on the top of an incline. It rolls down the incline without slipping in exactly 50. The radius of the sphere is

(a.) $\left(\frac{5}{\pi}\right)$ cm
(b.) $\left(\frac{5}{\pi}\right)$ m
(c.) $\left(\frac{10}{\pi}\right)$ cm
(d.) 10 cm

93.



A set of concentric circles of integer radii 1, 2, ..., N is shown in the figure above. An ant starts at point A_1 , goes round the first circle, returns to A_1 , moves to A_2 , goes round the second circle, returns to A_2 , moves to A_3 and repeats this until it reaches A_N . The distance covered by the ant is

(a.) $N(N+1)\pi$
(b.) $2N\pi + N$
(c.) $\pi(N+1)N + N - 1$
(d.) $\pi(N-1)N + N - 1$

94.



The figure above shows an infinite series of triangles, in which $r_1 > r_2 > r_3 \dots$. What is the total length of the solid line segments in the figure?

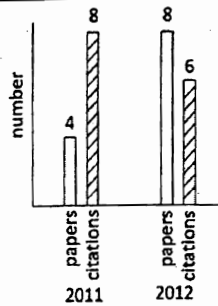
(a.) $\frac{r_1}{r_2} + \frac{r_2}{r_3} + \dots$
(b.) $\frac{r_1}{r_1 - r_2}$
(c.) $\frac{r_1^2}{r_1 + r_2}$
(d.) $\frac{r_1 - r_2}{r_1^2}$

95.

If a_i , b_i and c_i are distinct, how many terms will the expansion of the product $(a_1 + a_2 + a_3)(b_1 + b_2 + b_3 + b_4)(c_1 + c_2 + c_3 + c_4 + c_5)$ contain?

(a.) 12
(b.) 30
(c.) 23
(d.) 60

96.



The above plot depicts the number of research publications of a scientist along with the number of citations. Which of the following statements is **not** correct?

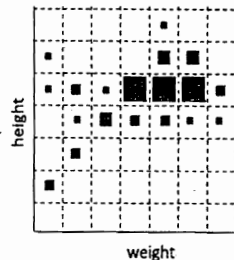
- (a.) In the year 2012, 50% more papers were published but citations decreased by 25%.
- (b.) In the year 2012, 100% more papers were published but citations were 75% of the number of papers in that year.
- (c.) The papers published in year 2011 is only 33.33% of the total number of papers in both years.
- (d.) The total number of citations for both years is 16.66% more than the total number of papers.

97.

The next number of the sequence 1, 5, 14, 30, 55, ... is

- (a.) 85
- (b.) 90
- (c.) 91
- (d.) 95

98.



The distribution of heights and weights in a population is shown above in a 2- parameter scatter plot. The size of the square is proportional to the number of persons having a particular combination of weight and height. Which statement best describes the trend in the population?

- (a.) Height and weight are strongly correlated.
- (b.) Height and weight are anticorrelated.
- (c.) Large heights do not imply proportionately large weights.
- (d.) Height and weight are independent characteristics.

99.

What is the maximum sum of the numbers of Saturdays and Sundays in a leap year?

- (a.) 104
- (b.) 105
- (c.) 106
- (d.) 107

100.

Two trains of lengths 150 m and 250 m pass each other with constant speeds on parallel tracks in opposite directions. The drivers and guards are at the extremities of the trains. The time gap between the drivers passing each other and first driver-guard pair passing each other is 30 s. How much later will the other driver-guard pair pass by?

- (a.) 10 s
- (b.) 20 s
- (c.) 30 s
- (d.) 50 s

101.

What is the angle between the minute and hour hands of a clock at 7 : 35?

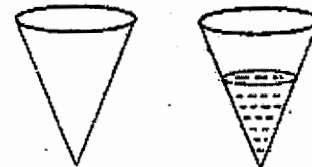
- (a.) 0°
- (b.) 17.5°
- (c.) 19.5°
- (d.) 20°

102.

A stream of ants go from point A to point B and return to A along the same path. All the ants move at a constant speed and from any given point 2 ants pass per second one way. It takes 1 minute for an ant to go from A to B. How many returning ants will an ant meet in its journey from A to B?

- (a.) 120
- (b.) 60
- (c.) 240
- (d.) 180

103.



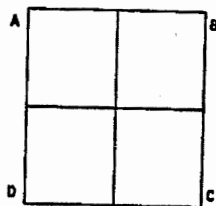
The capacity of the conical vessel shown above is V. It is filled with water upto half its height. The volume of water in the vessel is

- (a.) $\frac{V}{2}$
- (b.) $\frac{V}{4}$
- (c.) $\frac{V}{8}$
- (d.) $\frac{V}{16}$

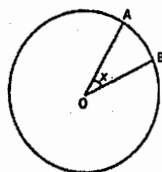
104. A large tank filled with water is to be emptied by removing half of the water present in it everyday. After how many days will there be closest to 10% water left in the tank?
 (a.) One
 (b.) Two
 (c.) Three
 (d.) Four
105. n is a natural number. If n^5 is odd, which of the following is true?
 A. n is odd
 B. n^3 is odd
 C. n^4 is even
 (a.) A only
 (b.) B only
 (c.) C only
 (d.) A and B only
106. Suppose you expand the product $(x_1 + y_1)(x_2 + y_2) \dots (x_{20} + y_{20})$. How many terms will have only one x and rest y 's?
 (a.) 1
 (b.) 5
 (c.) 10
 (d.) 20
107. A 16.2 m long wooden log has a uniform diameter of 2 m. To what length of log should be cut to obtain a piece of 22 m^3 volume?
 (a.) 3.5 m
 (b.) 7.0 m
 (c.) 14.0 m
 (d.) 22.0 m
108. What is the last digit of 7^{73} ?
 (a.) 7
 (b.) 9
 (c.) 3
 (d.) 1
109. A lucky man finds 6 pots of gold coins. He counts the coins in the first four pots to be 60, 30, 20 and 15, respectively. If there is a definite progression, what would be the numbers of coins in the next two pots?
 (a.) 10 and 5
 (b.) 4 and 2
 (c.) 15 and 15
 (d.) 12 and 10
110. A bee leaves its hive in the morning and after flying for 30 minutes due south reaches a garden and spends 5 minutes collecting honey. Then it flies for 40 minutes due west and collects honey in another garden for 10 minutes. Then it returns to the hive taking the shortest route. How long was the bee away from its hive? (Assume that the bee flies at constant speed)
 (a.) 85 min
 (b.) 155 min
 (c.) 135 min
 (d.) Less than 1 hour
111. Find the missing number:

 (a.) 1
 (b.) 0
 (c.) 2
 (d.) 3
112. In solving a quadratic equation of the form $x^2 + ax + b = 0$, one student took the wrong value of a and got the roots as 6 and 2; while another student took the wrong value of b and got the roots as 6 and 1. What are the correct values of a and b , respectively?
 (a.) 7 and 12
 (b.) 3 and 4
 (c.) -7 and 12
 (d.) 8 and 12
113. The distance between two oil rigs is 6 km. What will be the distance between these rigs in maps of 1 : 50000 and 1 : 5000 scales, respectively?
 (a.) 12 cm and 1.2 cm
 (b.) 2 cm and 12 cm
 (c.) 120 cm and 12 cm
 (d.) 12 cm and 120 cm
114. A bird perched at the top of a 12 m high tree sees a centipede moving towards the base of the tree from a distance equal to twice the height of the tree. The bird flies along a straight line to catch the centipede. If both move at the same speed, at what distance from the base of the tree will the centipede be picked up by the bird?
 (a.) 16 m
 (b.) 9 m
 (c.) 12 m
 (d.) 14 m

115. An ant goes from A to C in the figure crawling only on the lines and taking the least length of path. The number of ways in which it can do so is



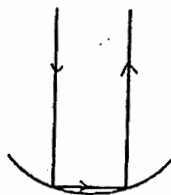
- (a.) 2
(b.) 4
(c.) 5
(d.) 6
116. A point is chosen at random from a circular disc shown below. What is the probability that the point lies in the sector OAB?



(Where $\angle AOB = x$ radians)

- (a.) $\frac{2x}{\pi}$
(b.) $\frac{x}{\pi}$
(c.) $\frac{x}{2\pi}$
(d.) $\frac{x}{4\pi}$

- 117.



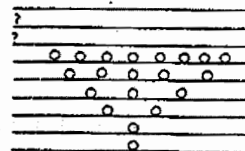
A ray of light, after getting reflected twice from a hemispherical mirror of radius R (see the above figure), emerges parallel to the incident ray. The separation of the original incident ray and the final reflected ray is

- (a.) R
(b.) $R\sqrt{2}$
(c.) $2R$
(d.) $R\sqrt{3}$

118. A king ordered that a golden crown be made for him from 8 kg of gold and 2 kg of silver. The goldsmith took away some amount of gold and replaced it by an equal amount of silver and the crown when made, weighed 10 kg. Archimedes knew that under water gold lost $1/20^{\text{th}}$ of its weight, while silver lost $1/10^{\text{th}}$. When the crown was weighed under water, it was 9.25 kg. How much gold was stolen by the goldsmith?

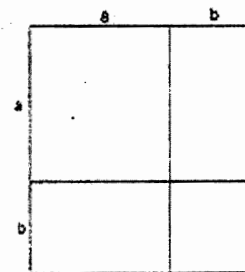
- (a.) 0.5 kg
(b.) 1 kg
(c.) 2 kg
(d.) 3 kg

119. In the figure below the numbers of circles in the blank row must be



- (a.) 12 and 20
(b.) 13 and 20
(c.) 13 and 21
(d.) 10 and 11

120. Which of the following is indicated by the accompanying diagram?



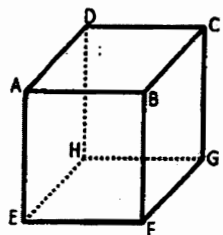
- (a.) $a + ab + ab^2 + \dots = a/(1-b)$ for $|b| < 1$
(b.) $a > b$ implies $a^3 > b^3$
(c.) $(a+b)^2 = a^2 + 2ab + b^2$
(d.) $a > b$ implies $-a < -b$

121. Which of the following numbers is the largest?

$2^{3^4}, 2^{4^3}, 3^{4^2}, 4^{2^3}, 4^{3^2}$

- (a.) 2^{3^4}
(b.) 3^{4^2}
(c.) 4^{3^2}
(d.) 4^{2^3}

122. The cube ABCDEFGH in the figure has each edge equal to a . The area of the triangle with vertices at A, C and F is



- (a.) $\frac{\sqrt{3}}{4} a^2$
(b.) $\frac{\sqrt{3}}{2} a^2$
(c.) $\sqrt{3} a^2$
(d.) $2\sqrt{3} a^2$
123. What is the number of distinct arrangements of the letters of the word UGCCSIR so that U and I cannot come together?

- (a.) 2520
(b.) 720
(c.) 1530
(d.) 1800
124. Suppose the sum of the even positive numbers is 21. What is the minimum possible value of the average of the square of these numbers?

- (a.) 63
(b.) 21
(c.) 9
(d.) 7

125. Let
$$A = \frac{1^{13} + 2^{13} + 3^{13} + \dots + 100^{13}}{100},$$

$$B = \frac{1^{13} + 3^{13} + 5^{13} + \dots + 99^{13}}{50},$$

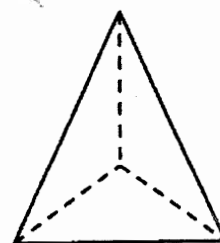
$$C = \frac{2^{13} + 4^{13} + 6^{13} + \dots + 100^{13}}{50}. \text{ Which of the following is true?}$$

- (a.) $B < C < A$
(b.) $A < B < C$
(c.) $B < A < C$
(d.) $C < A < B$

126. A circle of radius 5 units in the XY plane has its centre in the first quadrant, touches the x -axis and has a chord of length 6 units on the y -axis. The coordinates of its centre are

- (a.) (4, 6)
(b.) (3, 5)
(c.) (5, 4)
(d.) (4, 5)

127. A wire of length 6m is used to make a tetrahedron of each edge 1m, using only one strand of wire for each edge. The minimum number of times the wire has to be cut is



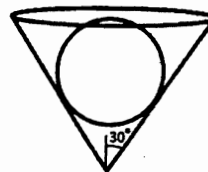
- (a.) 2
(b.) 3
(c.) 1
(d.) 0

128. If the sum of the next two terms of the series below is x , what is the value of $\log_2 x$?

2, -4, 8, -16, 32, -64, 128,

- (a.) 128
(b.) 10
(c.) 256
(d.) 8

129.



A conical vessel with semi-vertical angle 30° and height 10.5 cm has a thin lid. A sphere kept inside it touches the lid. The radius of the sphere is cm is

- (a.) 3.5
(b.) 2
(c.) 6.5
(d.) 7

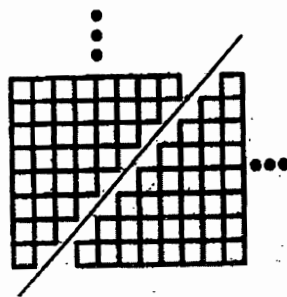
130. Amar, Akbar and Anthony are three friends, one of whom is a doctor, another is an engineer and the third is a professor. Amar is not a engineer. Akbar is the shortest. The tallest person is a doctor. The engineer's height is the geometric mean of the heights of the other two. Then which of the following is true?

(a.) Amar is a doctor and he is the tallest
(b.) Akbar is a professor and he is the tallest
(c.) Anthony is an engineer and he is shortest
(d.) Anthony is a doctor and he is the tallest

131. If 100 cats each 100 mice in 100 minutes, then how long will it take for 7 cats to catch 7 mice?

(a.) 100/7 minutes
(b.) 100 minutes
(c.) 49/100 minutes
(d.) 7 minutes

132. What does the diagram demonstrate?



(a.) $1+2+3+\dots+n = \frac{n \cdot (n+1)}{2}$
(b.) $1^2+2^2+3^2+\dots+n^2 = \frac{n \cdot (n+1) \cdot (2n+1)}{6}$
(c.) $1+3+\dots+(2n+1) = n^2$
(d.) $2^2+4^2+\dots+(2n)^2 = \frac{2n \cdot (n+1) \cdot (2n+1)}{3}$

133. Suppose there are socks of N different colours in a box. If you take out one sock at a time, what is the maximum number of socks that you have to take out before a matching pair is found? Assume that N is an even number.

(a.) N
(b.) $N+1$
(c.) $N-1$
(d.) $\frac{N}{2}$

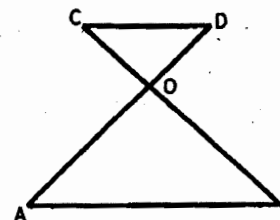
134. At what time after 4 O' clock, the hour and the minute hands will lie opposite to each other?

(a.) $4-50'+31''$
(b.) $4-52'-51''$
(c.) $4-53'-23''$
(d.) $4-54'-33''$

135. Which of the following curves just touches the x axis?

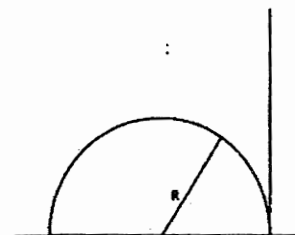
(a.) $y = x^2 - x + 1$
(b.) $y = x^2 - 2x + 2$
(c.) $y = x^2 - 10x + 25$
(d.) $y = x^2 - 7x + 12$

136. If AB is parallel to CD and $AO = 2OD$, then the area of triangle OAB is bigger than the area of triangle OCD by a factor of



(a.) 2
(b.) 3
(c.) 4
(d.) 8

137. A semi-circular arch of radius R has a vertical pole put on the ground together with one of its legs. A ant on the top of the arch finds the angular height of the tip of the pole to be 45° . The height of the pole is



(a.) $\sqrt{2} R$
(b.) $\sqrt{3} R$
(c.) $\sqrt{4} R$
(d.) $\sqrt{5} R$

138. Suppose we make N identical smaller spheres from a big sphere. The total surface area of the smaller spheres is X times the total surface area of the big sphere, where X is
- (a.) $\sqrt{N}$
 - (b.) 1
 - (c.) $N^{1/3}$
 - (d.) N^3
139. What is the next number in the sequence 24, 30, 33, 39, 51,
- (a.) 57
 - (b.) 69
 - (c.) 54
 - (d.) 81
140. Four lines are drawn on a plane with no two parallel and no three concurrent. Lines are drawn joining the points of intersection of the previous four lines. The number of new lines obtained this way is
- (a.) 3
 - (b.) 2
 - (c.) 12
 - (d.) 2